Building and evaluation of a PBPK model for Desipramine in CYP2D6 phenotype and activity-score groups

Version	**evaluation-fork-fix-OSP12.2**
based on *Model Snapshot* and *Evaluation Plan*	https://github.com/Open-Systems-Pharmacology/Desipramine-Model/releases/tag/vevaluation-fork-fix
OSP Version	12.2
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This evaluation report and the corresponding PK-Sim project file are filed at:

https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library/

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1 Introduction

Desipramine is a tricyclic antidepressant and a sensitive CYP2D6 substrate. Its pharmacokinetics are suitable for evaluating CYP2D6 phenotype- and activity-dependent changes in exposure.

This desipramine model is intended to describe plasma concentration-time profiles of desipramine and 2-hydroxydesipramine after intravenous and oral administration across CYP2D6 phenotype and activity-score groups.

The model was developed using physicochemical, *in vitro*, and clinical pharmacokinetic information for desipramine and 2-hydroxydesipramine. Clinical data included intravenous and oral administration and poor-, extensive-, normal-, and higher-activity CYP2D6 groups. This compound report includes only desipramine-only or control-arm pharmacokinetic data.

The presented model includes the following features:

- desipramine and 2-hydroxydesipramine as parent and metabolite compounds,
- CYP2D6-mediated 2-hydroxylation with activity-score-dependent k_{cat} values,
- residual hepatic clearance processes,
- renal filtration with a GFR fraction of 1,
- intravenous and oral applications used for clinical model building and verification.

2 Methods

2.1 Modeling strategy

The general concept of building a PBPK model has previously been described by Kuepfer et al. (Kuepfer 2016). Relevant information on anthropometric and physiological parameters in adults was gathered from the literature and incorporated into PK-Sim as default values for adult simulations (Willmann 2007).

The applied activity and variability of plasma proteins and active processes integrated into PK-Sim are described in the publicly available PK-Sim Ontogeny Database or otherwise referenced for the specific process.

The desipramine parent-metabolite PBPK model was reported by Rüdesheim 2025. The model was built using intravenous and oral desipramine data to inform distribution, absorption, CYP2D6-mediated metabolism, and residual elimination.

Clinical studies used for model building covered intravenous and oral desipramine administration in CYP2D6 poor-, extensive-, and fast-metabolizer groups. Verification simulations included independent oral studies and activity-score groups. The model therefore separates parent desipramine performance from 2-hydroxydesipramine performance where metabolite data are available.

The intravenous data support distribution and systemic clearance assumptions without confounding by oral absorption. Oral studies then support absorption and first-pass metabolism. This split is important for desipramine because the same model must describe both parent exposure and metabolite formation across CYP2D6 function groups.

The structural model contains CYP2D6-mediated 2-hydroxylation, residual hepatic clearance, passive renal filtration, and oral absorption. CYP2D6 poor-metabolizer activity is represented by zero CYP2D6 k_{cat}, while normal and higher activity-score groups use optimized CYP2D6 k_{cat} values.

The evaluated applications include intravenous infusion or bolus dosing and oral dosing. The major proteins and processes represented explicitly are CYP2D6, plasma protein binding, residual hepatic clearance, and passive renal filtration.

The evaluation therefore uses parent concentration-time profiles as the primary readout and metabolite profiles where available as supporting evidence for the CYP2D6 formation pathway. Model-building and verification profiles are separated to preserve the logic of fitting structural parameters first and then evaluating independent clinical scenarios.

Details about input data are provided in Section 2.2. Details about the structural model and assumptions are provided in Section 2.3.

2.2 Data used

In vitro and physicochemical data

The table below summarizes the drug-dependent inputs documented for the desipramine model (Table 1). Parameter values are listed for desipramine and 2-hydroxydesipramine where they are relevant for interpreting the parent-metabolite simulations.

Parameter	Unit	Value	Source	Description
Desipramine				
MW	g/mol	266.4	Kim 2019	Molecular weight of desipramine free base.
pK_a,base	-	2.84	Swain 2012	pKa of a basic ionization site.
pK_a,base	-	10.02	Wishart 2006	pKa of a basic ionization site.
Solubility	mg/L	214.29	Swain 2012	Aqueous solubility at pH 6.5.
logP	-	3.52	Optimized	Octanol-water partition coefficient of the neutral species.
f_u	%	14	Watanabe 2018	Fraction unbound in plasma.
$K_{m,CYP2D6}$	µmol/L	0.73	Ball 1997; Austin 2002	Unbound-incubation-corrected Michaelis constant for 2-hydroxylation.
$k_{cat,CYP2D6}$, NM	1/min	5.03	Optimized	Catalytic rate constant for 2-hydroxylation in normal metabolizers.
CL_{hep}	1/min	0.54	Optimized	Unspecific hepatic clearance.
P_{int}	cm/min	1.14E-05	Optimized	Specific transcellular intestinal permeability.
Oral formulation	-	Solution	Assumed	Formulation used for oral predictions.
2-Hydroxydesipramine				
MW	g/mol	282.4	Kim 2019	Molecular weight.
pK_a,base	-	4.51	Swain 2012	pKa of a basic ionization site.
pK_a,base	-	9.90	Swain 2012	pKa of a basic ionization site.
pK_a,acid	-	10.63	Swain 2012	pKa of an acidic ionization site.
Solubility	mg/L	282.39	Swain 2012	Aqueous solubility at pH 6.5.

Parameter	Unit	Value	Source	Description
logP	-	2.33	Optimized	Octanol-water partition coefficient of the neutral species.
f_u	%	12	Watanabe 2018	Fraction unbound in plasma.
CL_{hep}	1/min	9.93	Optimized	Unspecific hepatic clearance.
P_{int}	cm/min	3.19E-05	Calculated	Specific transcellular intestinal permeability.
Shared distribution and elimination parameters				
GFR fraction	-	1.00	Assumed	Fraction used to scale passive glomerular filtration.
Partition coefficients	-	Rodgers and Rowland	Calculated	Tissue-to-plasma partition coefficients calculated with the Rodgers and Rowland method (Rodgers 2005, Rodgers 2006).
Cellular permeabilities	cm/min	0.27 / 0.01	Calculated	Cellular permeabilities for desipramine and 2-hydroxydesipramine calculated with the PK-Sim Standard method (OSP Documentation 2018).

Table 1: Drug-dependent physicochemical, distribution, metabolism, elimination, and formulation parameters used in the final desipramine model.

The CYP2D6 characterization-specific catalytic rate constants used in the model are listed below.

CYP2D6 characterization	k_{cat} [1/min]	Parameter origin
AS = 0 (PM)	0.00	Assumed
EM (model default)	5.03	Optimized
AS = 2 (NM)	9.86	Optimized
AS = 2.5 (UM)	28.88	Optimized

Table 1a: CYP2D6 characterization-specific k_{cat} values for desipramine 2-hydroxylation reported by Rüdesheim 2025. Parenthetical phenotypes for AS-coded rows use the current CPIC CYP2D6 activity score-to-phenotype mapping (Moore 2026). AS: activity score; EM: extensive metabolizer; NM: normal metabolizer; PM: poor metabolizer; UM: ultrarapid metabolizer. EM profiles and profiles without reported CYP2D6 information use the EM k_{cat}. Rüdesheim 2025 labels this group NM. Fast EM profiles (Brøsen 1986, Brøsen 1988) use the AS = 2 k_{cat}. Rüdesheim 2025 assumed this correspondence because these studies report no genotype or activity score and define fast extensive

metabolizers by a particularly low sparteine metabolic ratio. Poor-metabolizer profiles use the AS = 0 value.

Clinical data

The evaluation uses desipramine and 2-hydroxydesipramine plasma concentration-time profiles from peripheral venous blood. The profiles used for model building and verification are listed in Table 2.

Source	Dose [mg] / schedule*	Age [years]	Weight [kg]	Sex	N	Form.	CYP2D6 characterization
Aarnoutse 2005+	50 mg, oral, single dose	26 (20-56)	84 (69-93)	Male	13	NR	EM
Bergmann 2001	100 mg, oral, single dose	26.8 (24-33)	NR	83% female	6	NR	AS = 2 (NM)
Bergmann 2001	100 mg, oral, single dose	36.7 (22-54)	NR	33% female	6	NR	AS = 2.5 (UM)
Bergstrom 1992	50 mg, oral, single dose	44 (35-53)	78.2 (60.3-92.8)	Male	6	NR	EM
Boni 2009+	50 mg, oral, single dose	33.6 ± 7.2	80.9 ± 11.2	4% female	26	NR	EM
Brøsen 1986+	100 mg, oral, single dose	24.1 (21-30)	NR	50% female	6	NR	fast EM
Brøsen 1986+	100 mg, oral, single dose	28.8 (23-34)	NR	50% female	6	NR	PM
Brøsen 1986+	100 mg, oral, single dose	28.3 (21-36)	NR	50% female	6	NR	EM
Brøsen 1988+	50 mg, intravenous infusion over 60 min	23.75 (22-27)	NR	50% female	4	Solution	fast EM
Brøsen 1988+	50 mg, intravenous infusion over 60 min	29 (24-35)	NR	33% female	3	Solution	PM
Brøsen 1988+	50 mg, intravenous infusion over 60 min	29.75 (22-37)	NR	75% female	4	Solution	EM
Brøsen 1989	100 mg, oral, single dose	27.7 (23-38)	NR	50% female	6	NR	EM
Brøsen 1993	100 mg, oral, single dose	22-24	NR	Male	4	NR	fast EM
Brøsen 1993	100 mg, oral, single dose	22-24	NR	Male	8	NR	PM
Brøsen 1993	100 mg, oral, single dose	22-24	NR	Male	5	NR	EM
Harris 2007+	50 mg, oral, single dose	31.5 ± 10.8	NR	53% female	17	NR	EM

Source	Dose [mg] / schedule*	Age [years]	Weight [kg]	Sex	N	Form.	CYP2D6 characterization
Hynes 2015	50 mg, oral, single dose	58.3 ± 9.9	84.3 ± 28.5	55% female	20	NR	EM
Madani 2002+	50 mg, oral, single dose	18-45	NR	NR	24	NR	EM
Nichols 2009	50 mg, oral, single dose	35.0 ± 7.7	79.9 ± 9.8	15% female	20	NR	EM
Nichols 2013	50 mg, oral, single dose	18-45	>60	NR	38	NR	EM
Patroneva 2008	50 mg, oral, single dose	18-55	>50	NR	20	NR	EM
Reese 2008	50 mg, oral, single dose	19-35	NR	Male	15	NR	EM
Sauer 2004	50 mg, oral, single dose	26-55	NR	50% female	22	NR	EM
Skinner 2003	50 mg, oral, single dose	42 (21-63)	71.4	56% female	16	NR	EM
Spina 1987+	25 mg, oral, single dose	27-51	NR	21% female	8	NR	EM
Spina 1987+	25 mg, oral, single dose	27-51	NR	21% female	6	NR	PM
Spina 1995	100 mg, oral, single dose	24-38	67-81	Male	6	NR	EM

Table 2: Clinical desipramine concentration-time profiles used for model building and verification. *: Single oral dose unless otherwise specified; AS: activity score; EM: extensive metabolizer; fast EM: fast extensive metabolizer; NM: normal metabolizer; NR: not reported; PM: poor metabolizer; PT: predicted phenotype; UM: ultrarapid metabolizer; +: data used for model building. Parenthetical PTs for AS-coded rows use the current CPIC CYP2D6 activity score-to-phenotype mapping (Moore 2026). EM is the model default when study-specific CYP2D6 information is not available.

Hynes 2015 studied cancer patients. The simulation uses a healthy adult individual. The age range of the Reese 2008 subjects is taken from the description of the same study in the Wellbutrin XL prescribing information (Wellbutrin XL label).

2.3 Model parameters and assumptions

Absorption

The model includes intravenous and oral desipramine applications. Intravenous simulations do not require an absorption process. Oral desipramine simulations are represented with a solution-type formulation and compound-specific intestinal permeability.

The `Specific intestinal permeability` for desipramine was optimized using oral concentration-time data. The 2-hydroxydesipramine intestinal permeability was calculated and is relevant for metabolite simulations where the compound is present as a separate building block.

The intravenous simulations provide information for systemic disposition without absorption. Oral simulations use the same systemic disposition assumptions and add the intestinal permeability term. This allows the oral absorption parameter to be interpreted separately from the CYP2D6 and residual clearance parameters.

No phenotype-specific absorption parameter was introduced. Differences between CYP2D6 groups are represented through metabolism, while the same oral absorption assumptions are applied across adult simulations.

Distribution

Desipramine plasma protein binding was represented by a fraction unbound of 14% as summarized in Section 2.2. Lipophilicity was optimized and is an important parameter for the parent volume of distribution.

Partition coefficients were calculated with the Rodgers and Rowland method (Rodgers 2005, Rodgers 2006). Cellular permeabilities were calculated with the PK-Sim Standard method (OSP Documentation 2018). Because the model includes both parent and metabolite concentration-time profiles, distribution assumptions affect the interpretation of desipramine and 2-hydroxydesipramine profiles separately.

The model uses compound-specific distribution properties for desipramine and 2-hydroxydesipramine. Parent distribution is particularly important because desipramine is extensively distributed and the concentration-time profile after intravenous administration constrains this part of the model. Metabolite distribution is interpreted more cautiously because metabolite profiles are less comprehensive than parent profiles.

Distribution was kept independent of CYP2D6 status. CYP2D6 poor-, extensive-, and higher-activity groups differ through formation clearance, while the same adult physiological and binding assumptions are applied.

Metabolism and elimination

One explicit metabolic conversion and residual elimination pathways are represented in the model.

- CYP2D6

Desipramine is converted to 2-hydroxydesipramine by CYP2D6. CYP2D6 K_m is held constant across CYP2D6 groups. Poor-metabolizer k_{cat} is set to zero, while normal and higher activity-score k_{cat} values were optimized.

This structure assigns CYP2D6 phenotype and activity-score effects to catalytic capacity rather than to affinity.

- Residual elimination

The model includes residual hepatic clearance for desipramine and unspecific hepatic clearance for 2-hydroxydesipramine. Both parent and metabolite include passive renal filtration with a `GFR fraction` of 1.

Residual hepatic clearance accounts for desipramine elimination not explicitly assigned to CYP2D6 2-hydroxylation. The metabolite unspecific hepatic clearance represents downstream elimination of 2-hydroxydesipramine. Renal filtration is implemented as a passive process using adult renal physiology and the fraction unbound of the respective compound.

3 Results and Discussion

The PBPK model for desipramine was developed and evaluated with clinical pharmacokinetic data after intravenous and oral administration. The evaluation covers desipramine and 2-hydroxydesipramine plasma concentration-time profiles and CYP2D6 poor-, extensive-, normal-, and higher-activity groups. It includes compound-only or control-arm profiles.

The model-building data informed systemic disposition, oral absorption, CYP2D6-mediated 2-hydroxylation, residual hepatic clearance, and metabolite elimination. Intravenous and oral phenotype-stratified data from Brøsen 1988, Brøsen 1986, and Spina 1987 supported CYP2D6 phenotype-dependent clearance. Oral studies with parent and metabolite measurements, including Boni 2009, supported the parent-metabolite structure. Additional oral data from Aarnoutse 2005, Harris 2007, and Madani 2002 supported adult oral exposure after 50 mg dosing.

Verification used independent studies across CYP2D6 activity groups, including compound-only or control-arm data from Bergmann 2001, Hynes 2015, Bergstrom 1992, Nichols 2013, Patroneva 2008, Skinner 2003, and Spina 1995. These data test whether the same model can describe adult oral desipramine exposure without study-specific refitting.

The model quantifies CYP2D6-mediated 2-hydroxylation, residual hepatic clearance of desipramine, unspecific hepatic clearance of 2-hydroxydesipramine, and passive glomerular filtration. CYP2D6 k_{cat} values represent the major phenotype and activity-score dependency. The metabolite pathway is important because adequate parent prediction alone would not demonstrate correct description of 2-hydroxydesipramine exposure.

The next sections show:

1. the final model input parameters for the building blocks: Section 3.1.
2. the overall goodness of fit: Section 3.2.
3. simulated vs. observed concentration-time profiles for the clinical studies used for model building and for model verification: Section 3.3.

The concentration-time profiles should be interpreted by dose route, analyte, and CYP2D6 activity group. Intravenous data mainly test systemic distribution and clearance. Oral data additionally test absorption and first-pass metabolism. Poor-metabolizer profiles are simulated without CYP2D6-mediated 2-hydroxylation, while normal- and higher-activity groups test the activity-score-dependent k_{cat} values.

The concentration-time profiles and goodness-of-fit diagnostics characterize model performance within the evaluated intravenous and oral dose range, analytes, and CYP2D6 groups. Unrepresented populations and tissue-specific pharmacodynamic endpoints remain outside the evidence evaluated in this report.

3.1 Desipramine final input parameters

The following tables summarize the final input parameters for desipramine and 2-hydroxydesipramine. Interpret optimized parameters together with the sources in Section 2.2 and the assumptions in Section 2.3.

Compound: Desipramine

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	214.29 mg/l	Database-Other-chemicalize.com	Predicted	True
Reference pH	6.5	Database-Other-chemicalize.com	Predicted	True
Lipophilicity	3.5204587578 Log Units	Parameter Identification-Parameter Identification-Optimized	Optimized	True
Lipophilicity	3.896 Log Units	Database-Other-chemicalize.com	Predicted	False
Fraction unbound (plasma, reference value)	0.135	Publication-Other-Predicted, Watanabe et al. 2018	Predicted	True
Specific intestinal permeability (transcellular)	1.14E-05 cm/min	Parameter Identification-Parameter Identification-Optimized	Optimized	True
Is small molecule	Yes			
Molecular weight	266.388 g/mol	Database-Other-PubChem (Kim et al. 2019)		
Plasma protein binding partner	α1-acid glycoprotein			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	PK-Sim Standard

Processes

Systemic Process: Glomerular Filtration-Assumption

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	Other-Other-Assumption

Systemic Process: Total Hepatic Clearance-Assumption

Species: Human

Parameters

Name	Value	Value Origin
Fraction unbound (experiment)	0.14	Unknown-Unknown
Lipophilicity (experiment)	3.52 Log Units	
Plasma clearance	0 ml/min/kg	
Specific clearance	0.5395490378 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Ball et al. 1997 (2-Hydroxydesipramine)

Molecule: CYP2D6

Metabolite: 2-Hydroxydesipramine

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	68 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.732 µmol/l	Publication-In Vitro-Ball et al. 1997, corrected for fu,inc 12% (Austin et al. 2002)
kcat	5.03 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=0 (2-Hydroxydesipramine)

Molecule: CYP2D6

Metabolite: 2-Hydroxydesipramine

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.73 µmol/l	Publication-In Vitro-Ball et al. 1997, corrected for fu,inc 12% (Austin et al. 2002)

Metabolizing Enzyme: CYP2D6-AS=2 (2-Hydroxydesipramine)

Molecule: CYP2D6

Metabolite: 2-Hydroxydesipramine

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.732 µmol/l	Publication-In Vitro-Ball et al. 1997, corrected for fu,inc 12% (Austin et al. 2002)
kcat	9.86 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=2.5 (2-Hydroxydesipramine)

Molecule: CYP2D6

Metabolite: 2-Hydroxydesipramine

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.732 µmol/l	Publication-In Vitro-Ball et al. 1997, corrected for fu,inc 12% (Austin et al. 2002)
kcat	28.88 1/min	Parameter Identification-Parameter Identification-Optimized

Compound: 2-Hydroxydesipramine

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	282.39 mg/l	Database-Other-chemicalize.com	Predicted	True
Reference pH	6.5	Database-Other-chemicalize.com	Predicted	True
Lipophilicity	2.33 Log Units	Parameter Identification-Parameter Identification-Optimized	Optimized	True
Lipophilicity	3.173 Log Units	Database-Other-chemicalize.com	Predicted	False
Fraction unbound (plasma, reference value)	0.12	Publication-Other-Predicted, Watanabe et al. 2018	Predicted	True
Is small molecule	Yes			
Molecular weight	282.38 g/mol	Database-Other-PubChem (Kim et al. 2019)		
Plasma protein binding partner	Unknown			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	PK-Sim Standard

Processes

Systemic Process: Glomerular Filtration-Assumption

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	Other-Other-Assumption

Systemic Process: Total Hepatic Clearance-Assumption

Species: Human

Parameters

Name	Value	Value Origin
Fraction unbound (experiment)	0.12	
Lipophilicity (experiment)	2.53 Log Units	Unknown-Unknown
Plasma clearance	0 ml/min/kg	
Specific clearance	9.93 1/min	Parameter Identification-Parameter Identification-Optimized

3.2 Diagnostic plots

The goodness-of-fit diagnostics are separated by analyte. Each analyte has separate plots and a separate GMFE table.

3.2.1 Desipramine goodness-of-fit diagnostics

The goodness-of-fit diagnostic plots show desipramine plasma concentration data used in the model evaluation.

The first plot shows observed versus simulated plasma concentrations and the line of identity. The second plot shows residuals versus time. Each residual is the natural logarithm of the observed concentration minus the natural logarithm of the simulated concentration.

Table 3-1: GMFE for Desipramine goodness-of-fit diagnostics

Group	GMFE
Desipramine	1.32

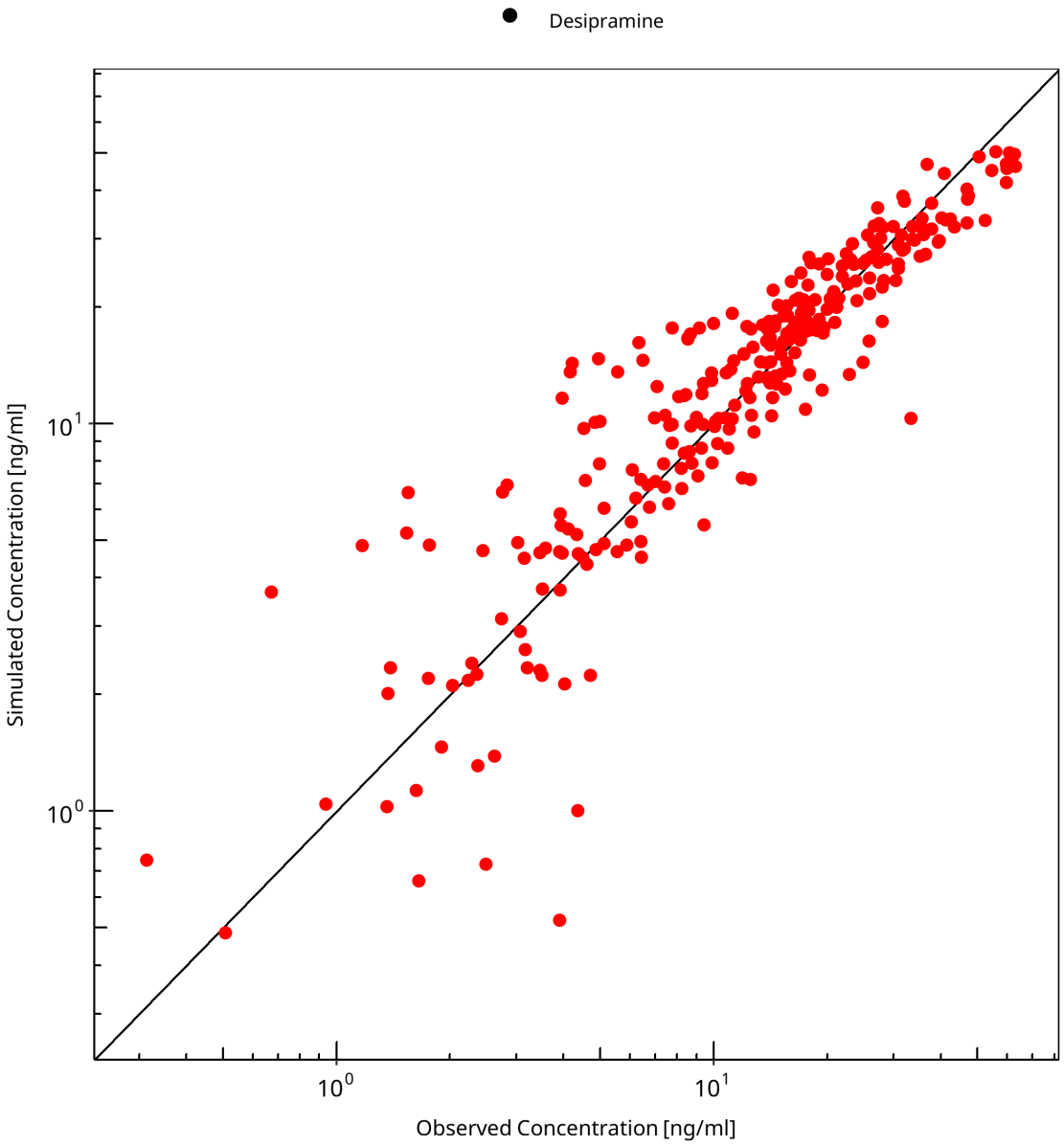


Figure 3-1: Desipramine goodness-of-fit diagnostics

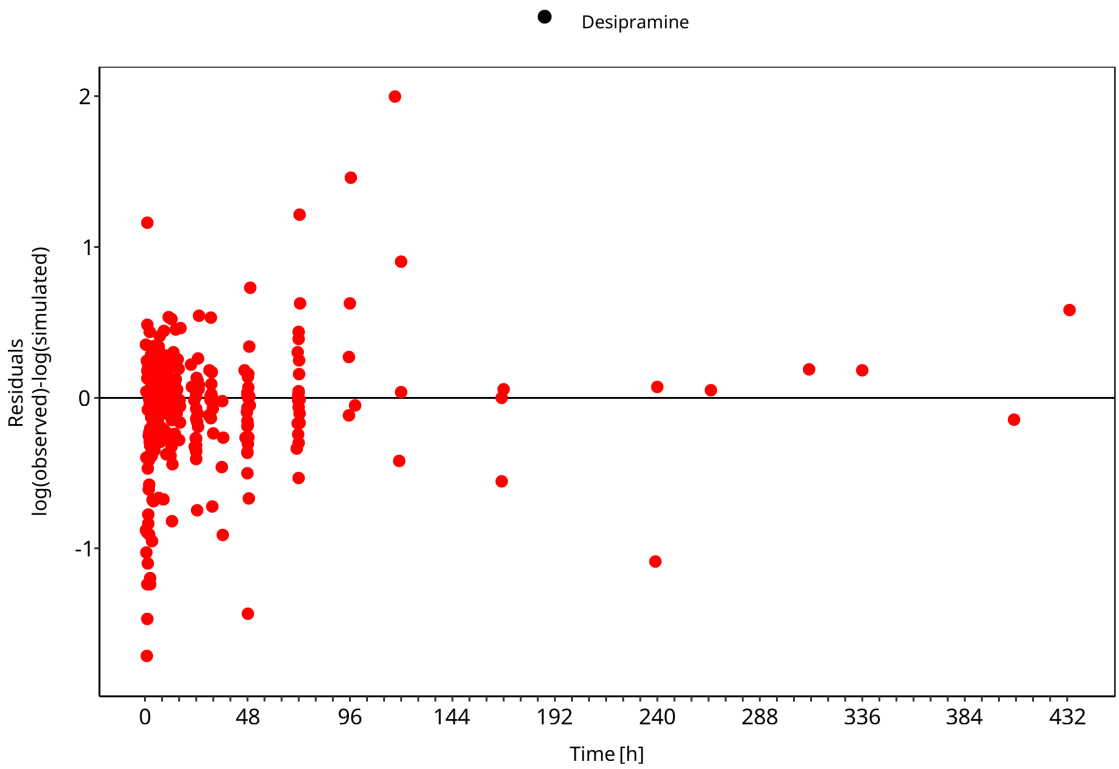


Figure 3-2: Desipramine goodness-of-fit diagnostics

3.2.2 2-Hydroxydesipramine goodness-of-fit diagnostics

The goodness-of-fit diagnostic plots show 2-hydroxydesipramine plasma concentration data used in the model evaluation.

The first plot shows observed versus simulated plasma concentrations and the line of identity. The second plot shows residuals versus time. Each residual is the natural logarithm of the observed concentration minus the natural logarithm of the simulated concentration.

Table 3-2: GMFE for 2-Hydroxydesipramine goodness-of-fit diagnostics

Group	GMFE
2-Hydroxydesipramine	1.29

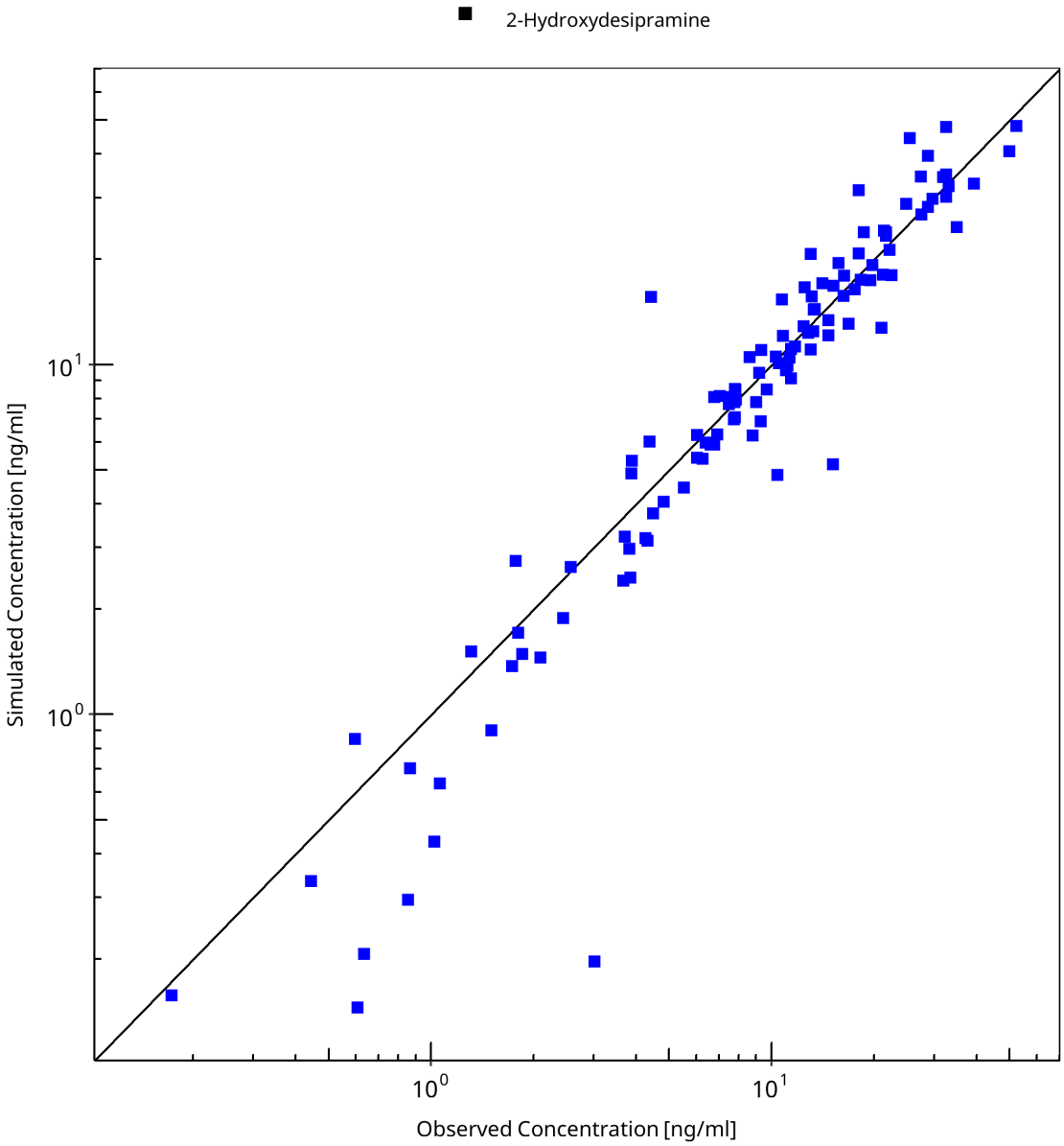


Figure 3-3: 2-Hydroxydesipramine goodness-of-fit diagnostics

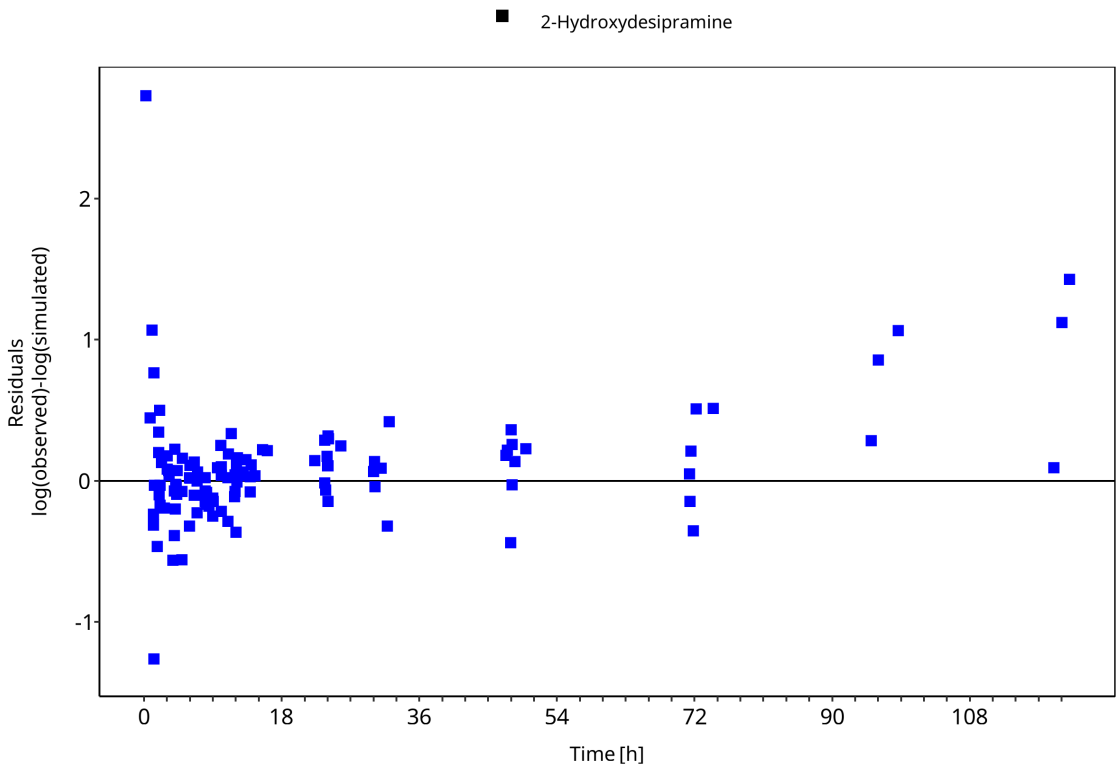


Figure 3-4: 2-Hydroxydesipramine goodness-of-fit diagnostics

3.3 Concentration-time profiles

Simulated versus observed concentration-time profiles of all studies listed in Table 2 are presented below.

3.3.1 Model building

The following model-building profiles show the clinical data used to establish parent and metabolite disposition, oral absorption, and CYP2D6-mediated 2-hydroxylation. Intravenous and oral data are shown separately where available because they inform different model components.

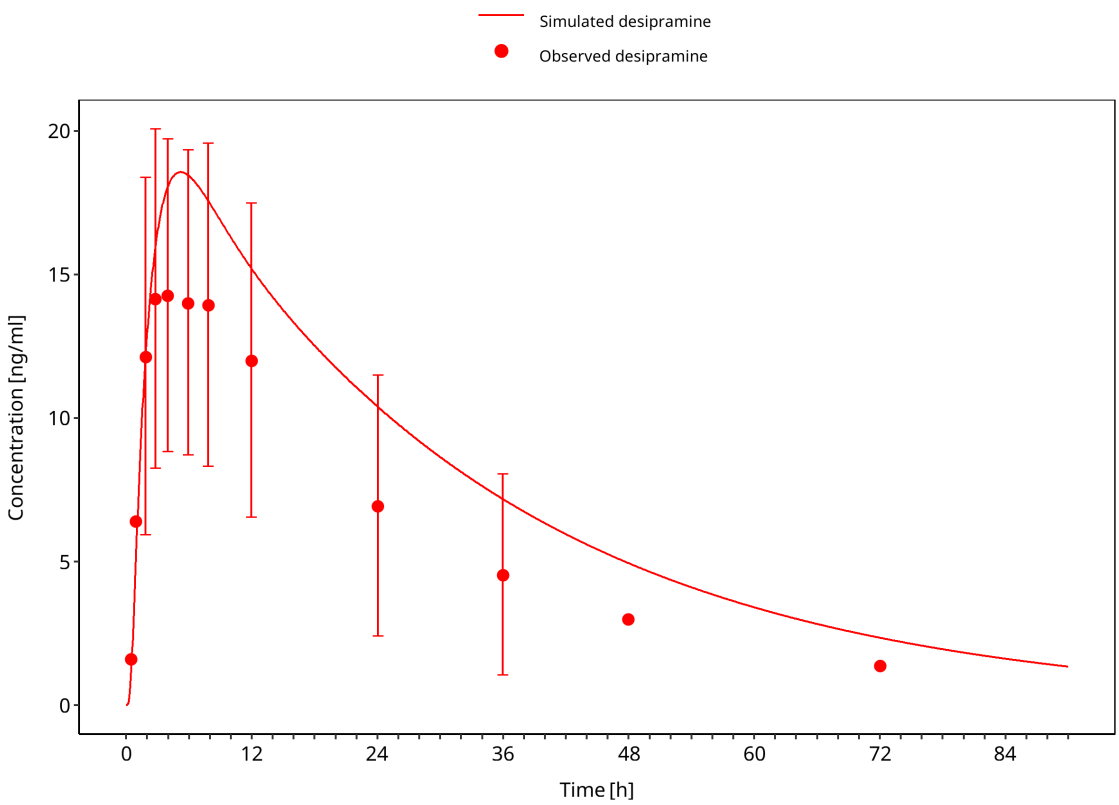


Figure 3-5: Aarnoutse 2005: desipramine hydrochloride, po, 50 mg, n=13 (EM)

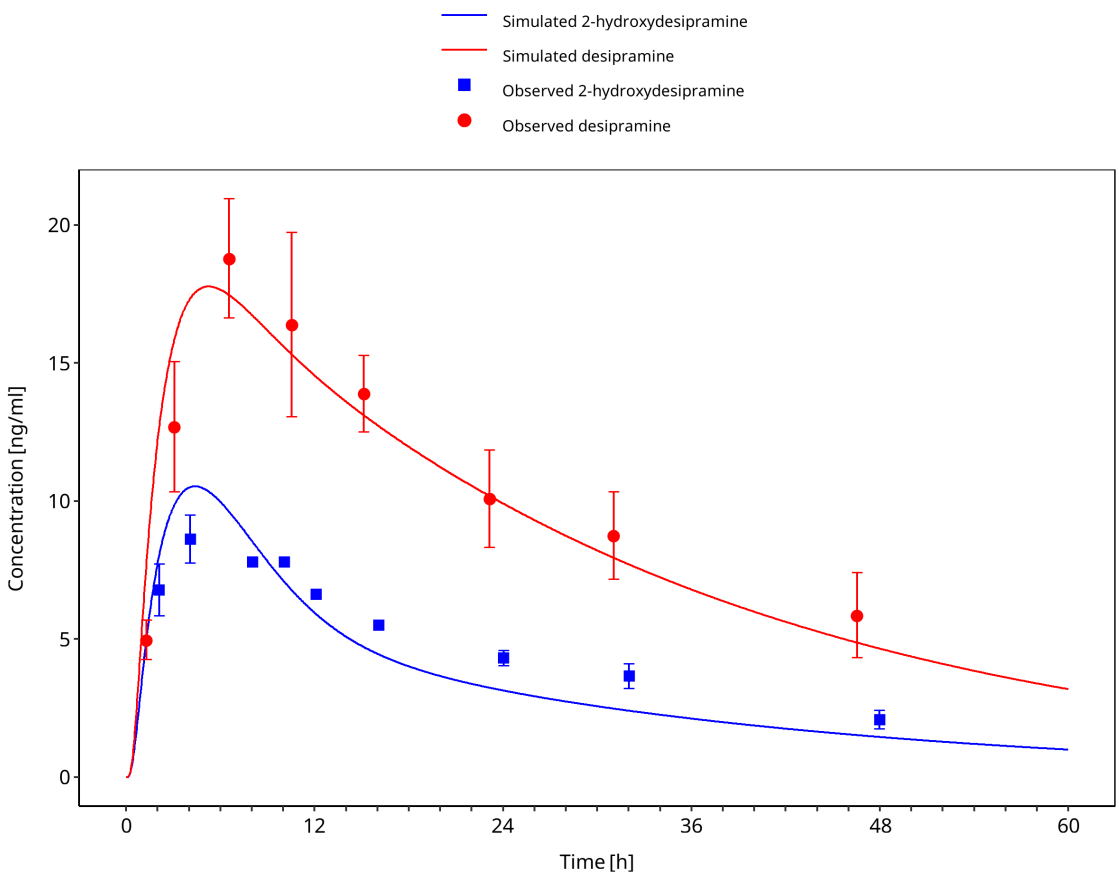


Figure 3-6: Boni 2009: desipramine hydrochloride, po, 50 mg, n=26 (EM)

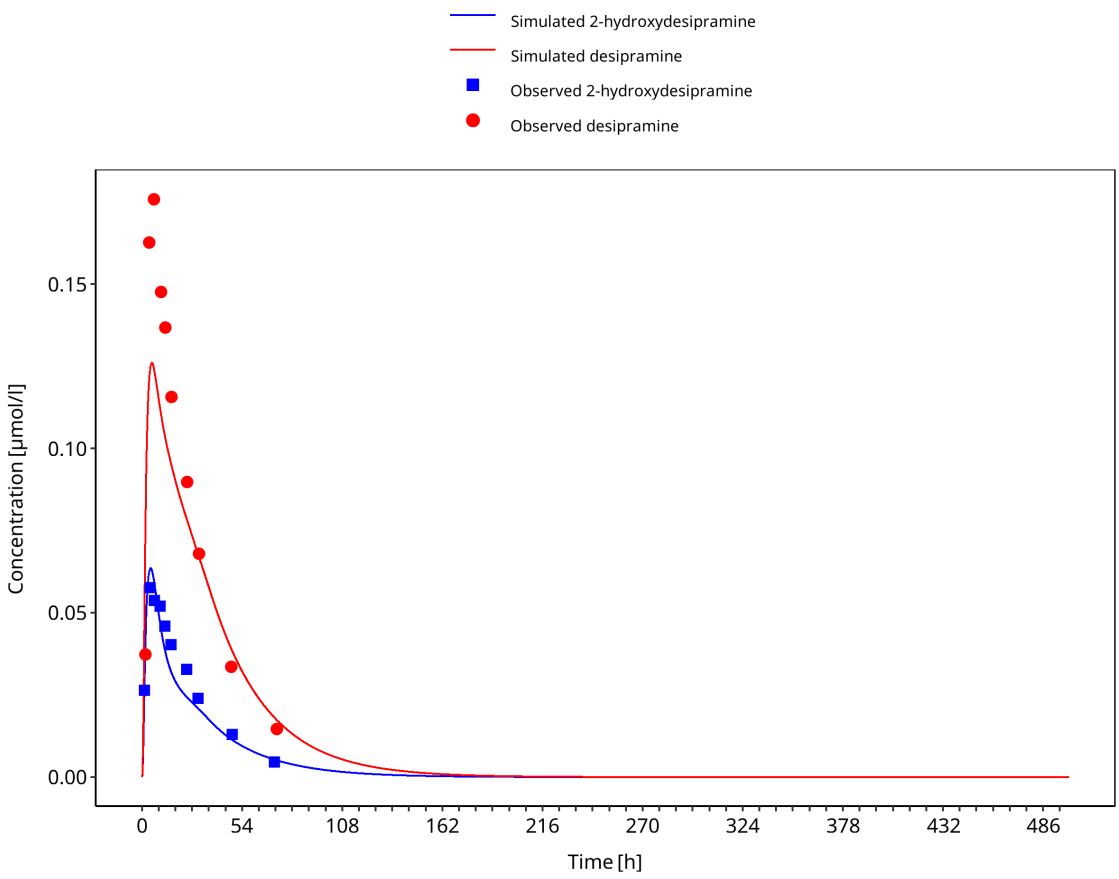


Figure 3-7: Brøsen 1986: desipramine hydrochloride, po, 100 mg, n=6 (EM)

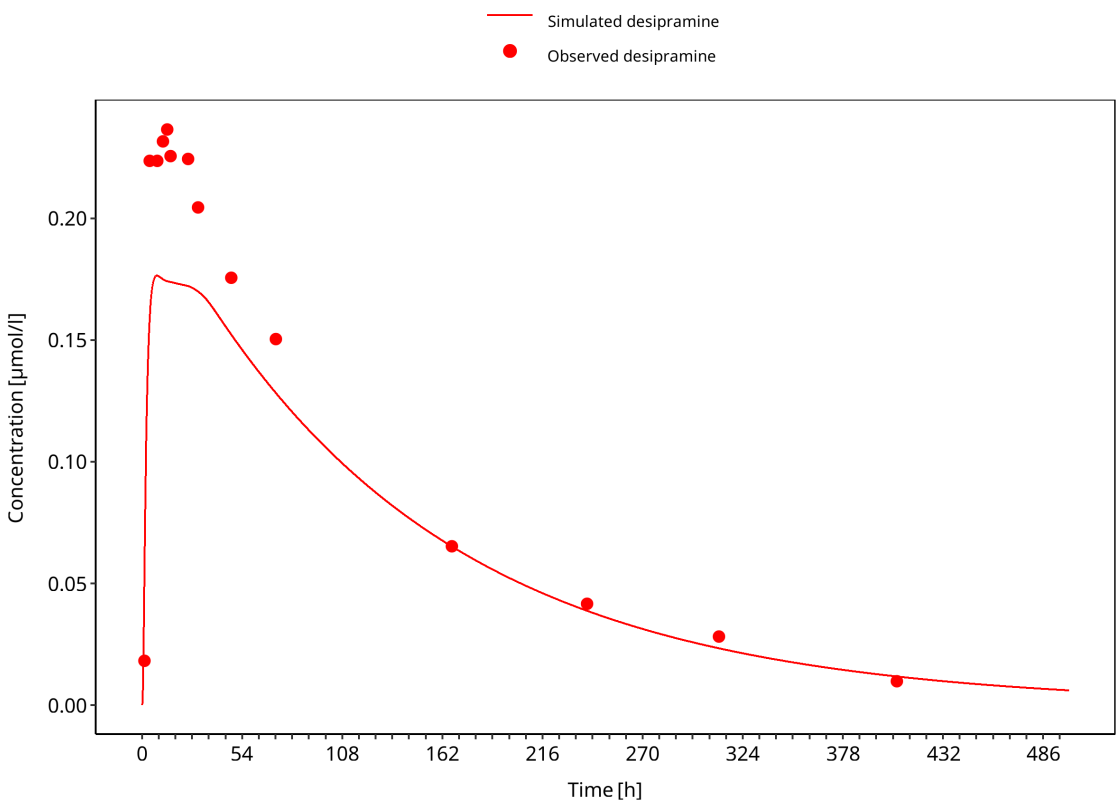


Figure 3-8: Brøsen 1986: desipramine hydrochloride, po, 100 mg, n=6 (PM)

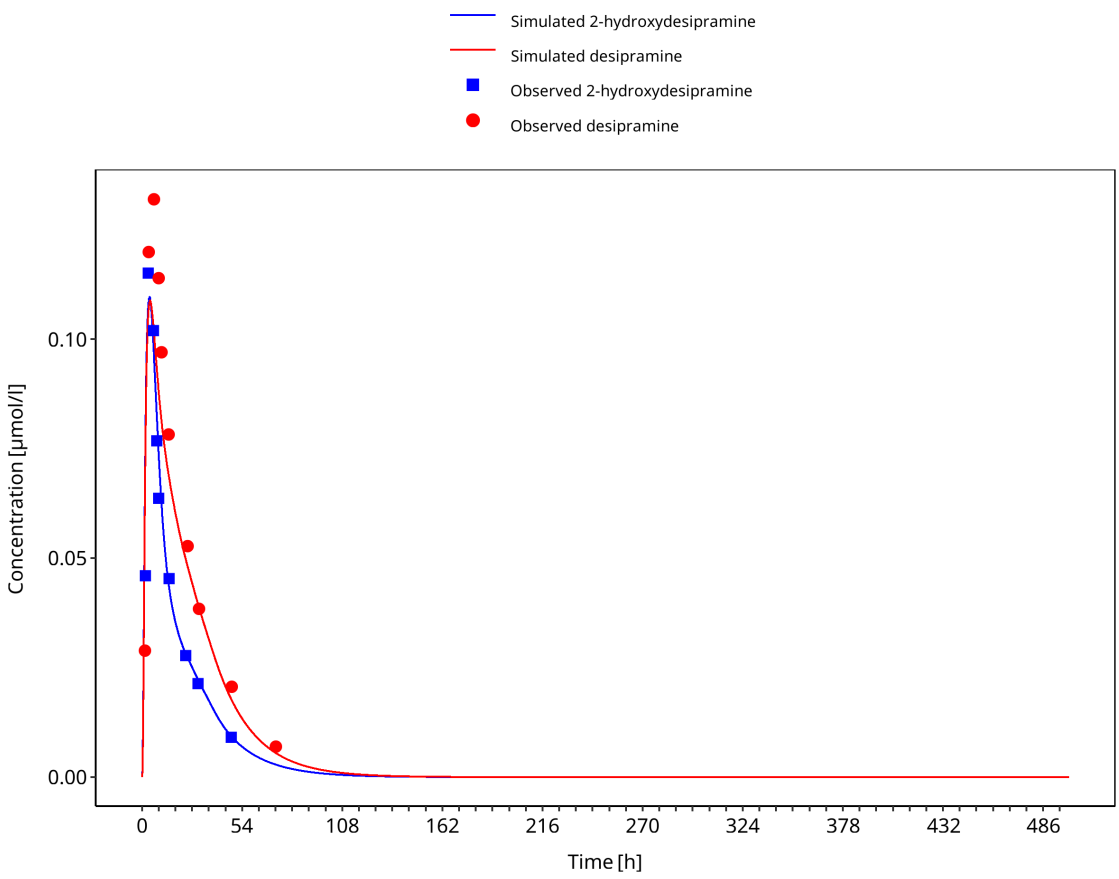


Figure 3-9: Brøsen 1986: desipramine hydrochloride, po, 100 mg, n=6 (fast EM)

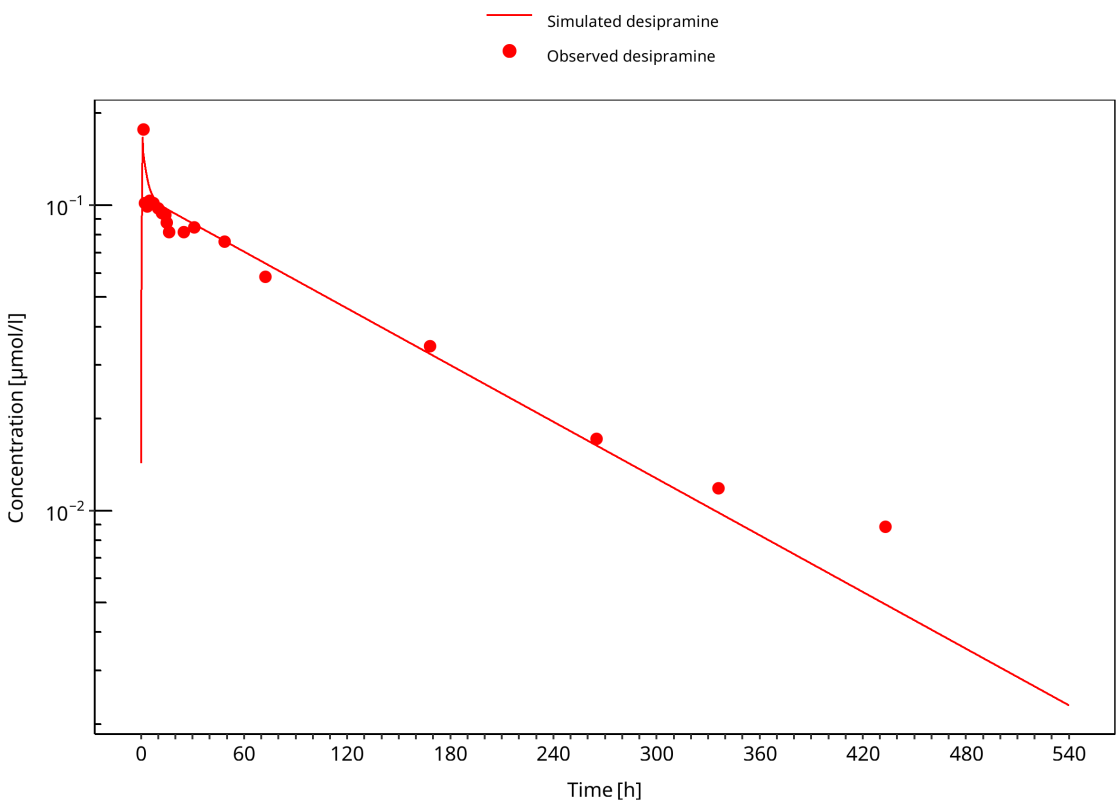


Figure 3-10: Brøsen 1988: desipramine hydrochloride, iv, 50 mg, n=3 (PM)

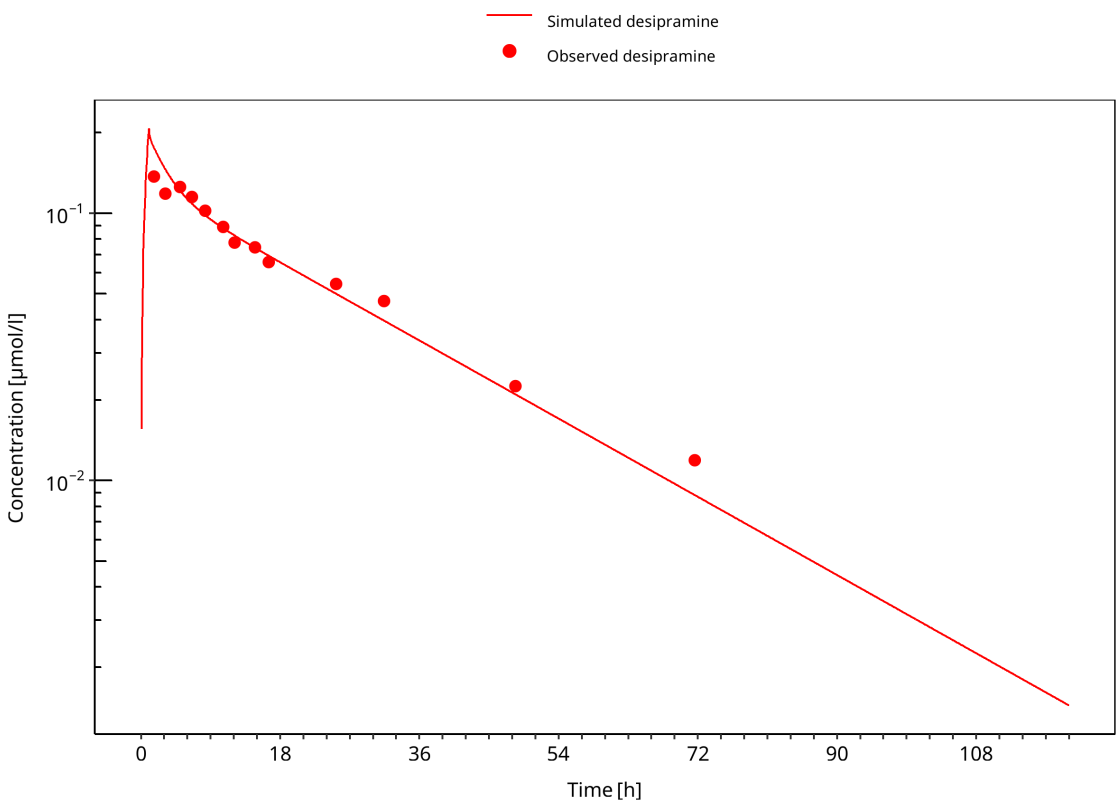


Figure 3-11: Brøsen 1988: desipramine hydrochloride, iv, 50 mg, n=4 (EM)

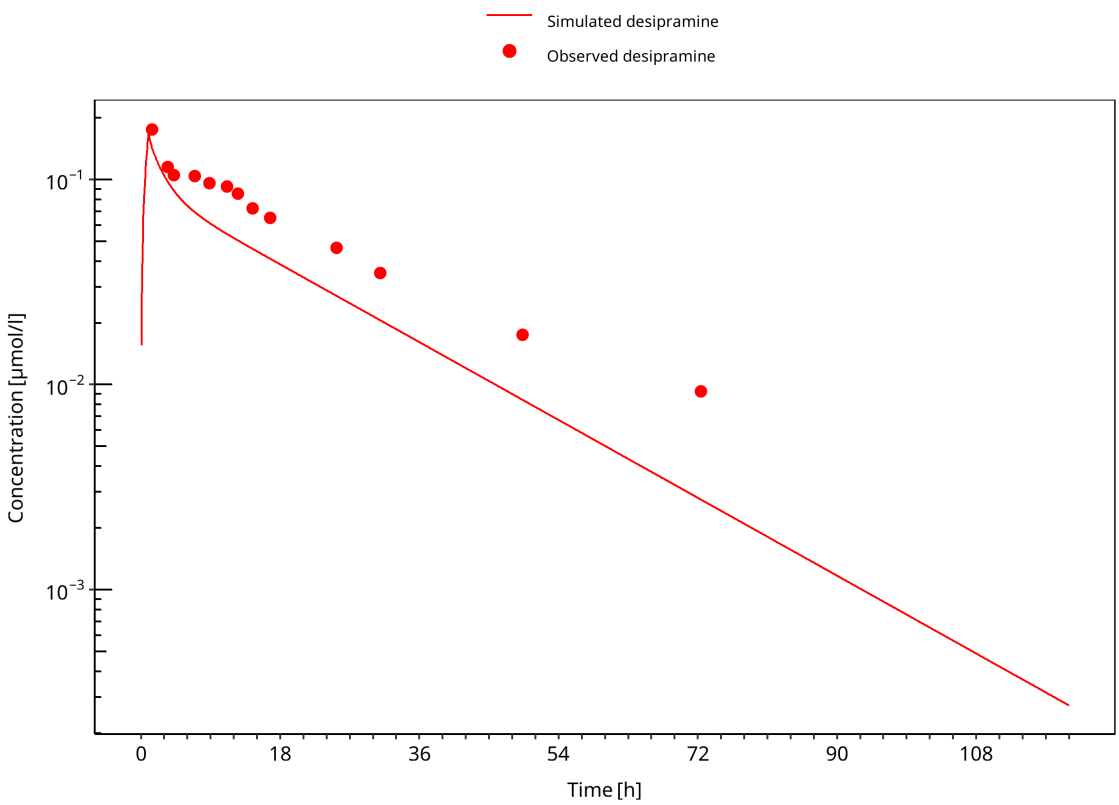


Figure 3-12: Brøsen 1988: desipramine hydrochloride, iv, 50 mg, n=4 (fast EM)

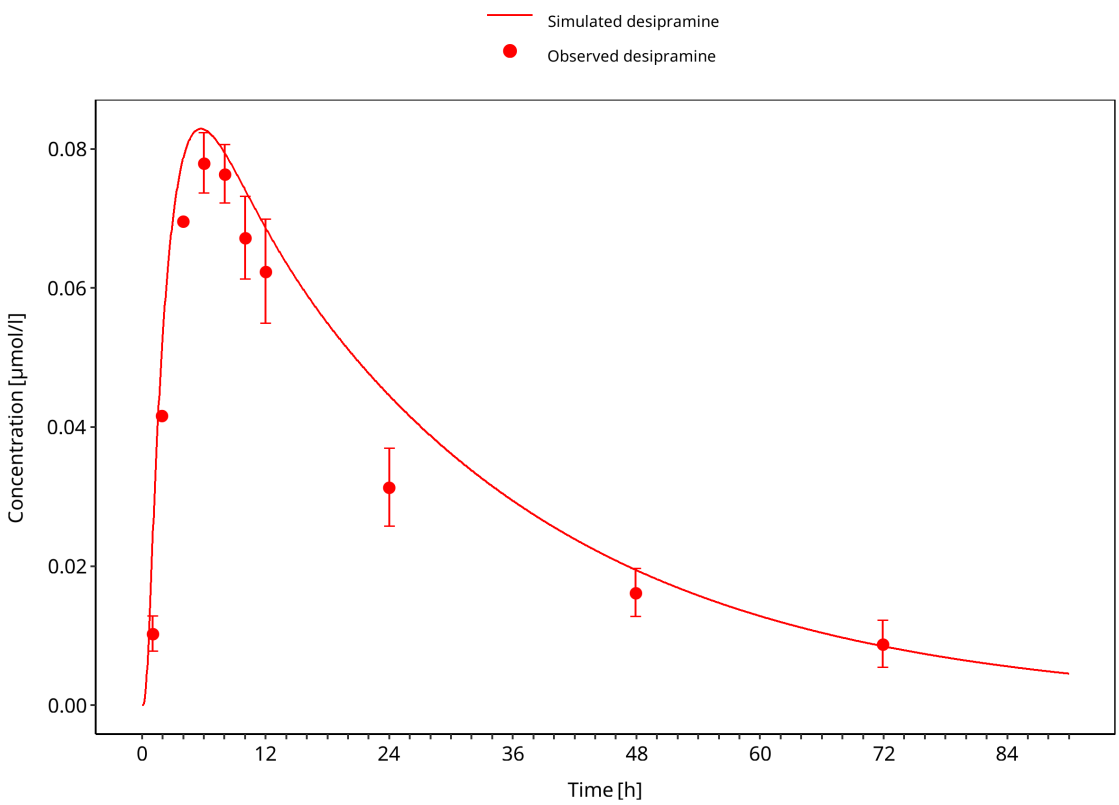


Figure 3-13: Harris 2007: desipramine hydrochloride, po, 50 mg, n=17 (EM)

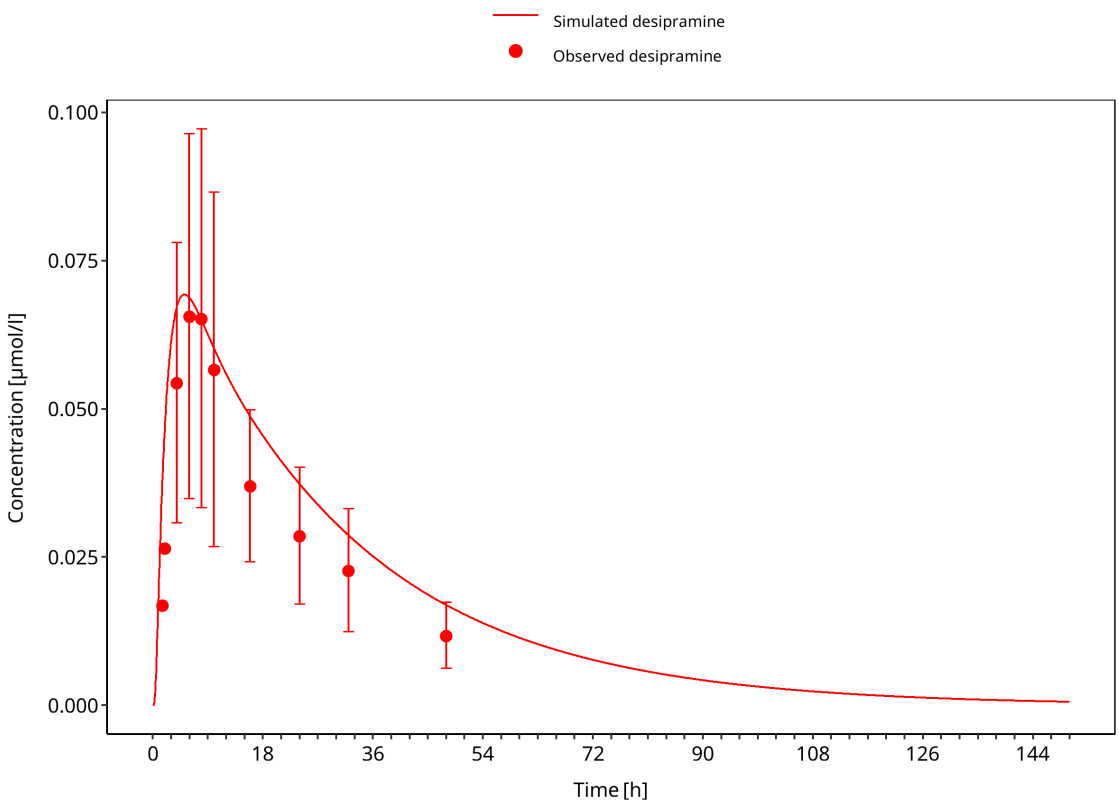


Figure 3-14: Madani 2002: desipramine hydrochloride, po, 50 mg, n=24 (EM)

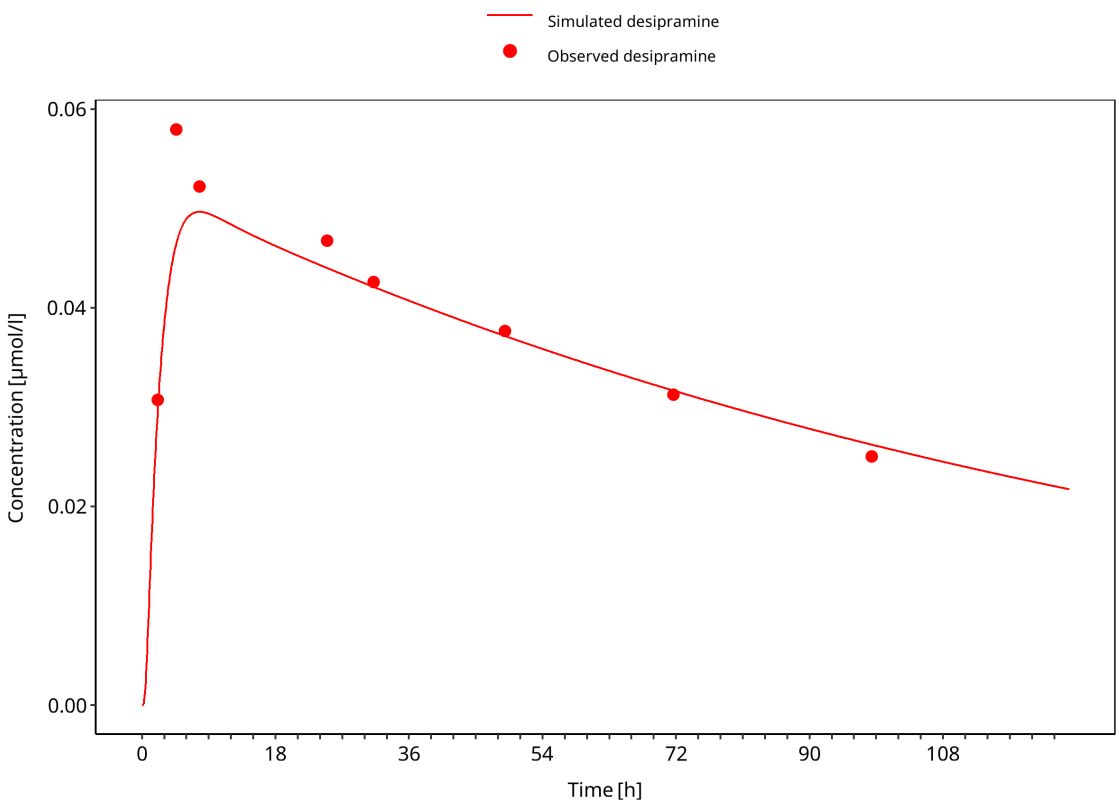


Figure 3-15: Spina 1987: desipramine hydrochloride, po, 25 mg, n=6 (PM)

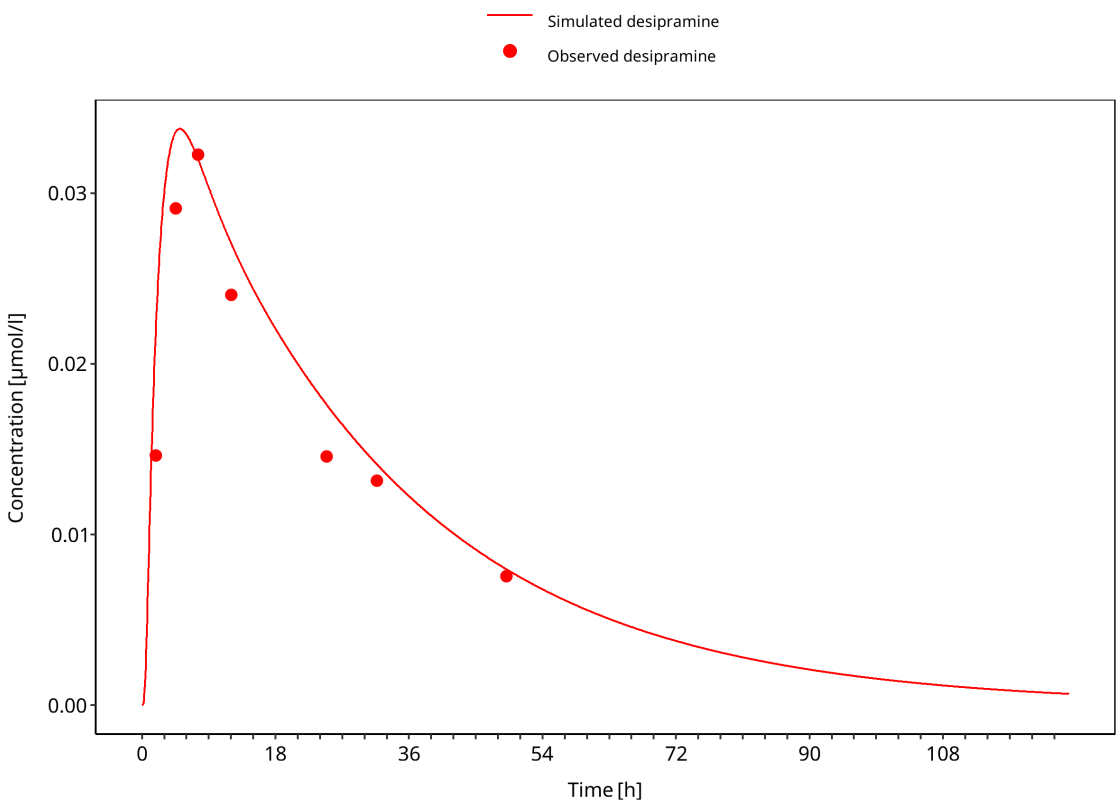


Figure 3-16: Spina 1987: desipramine hydrochloride, po, 25 mg, n=8 (EM)

3.3.2 Model verification

Simulated versus observed concentration-time profiles for model verification studies are presented below.

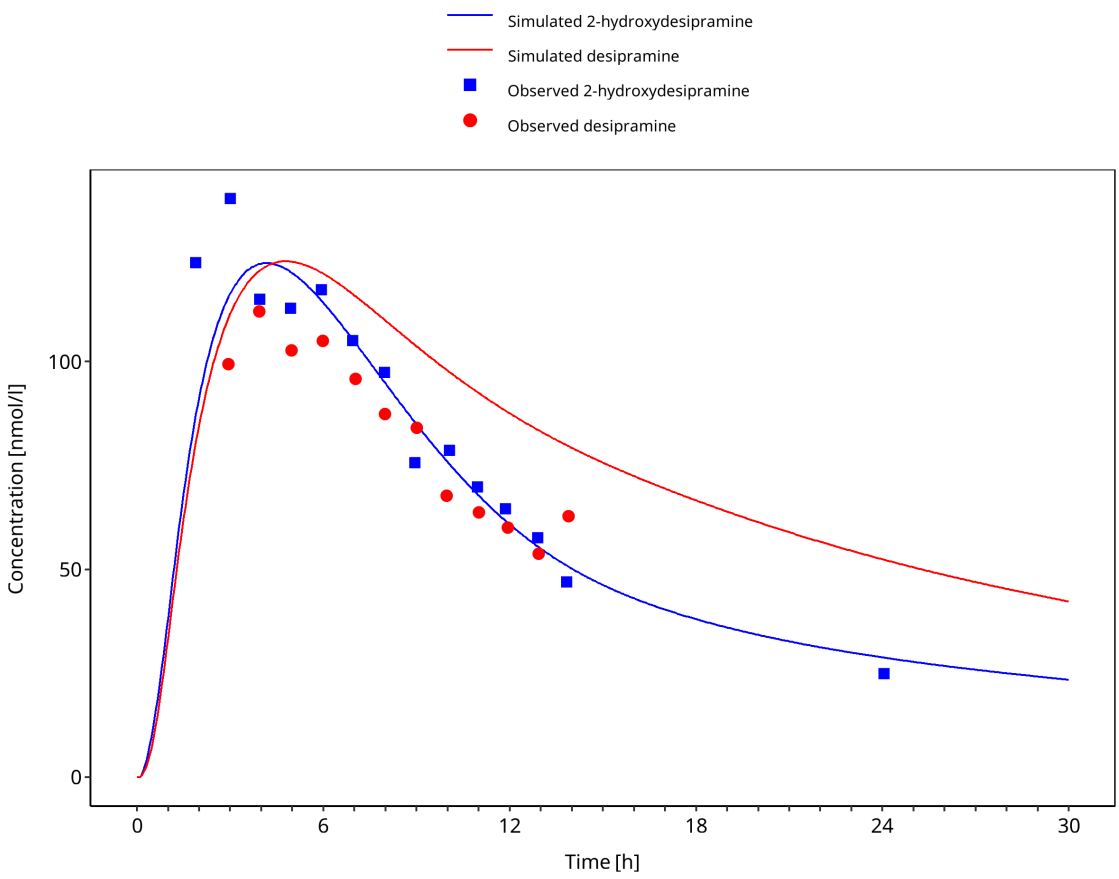


Figure 3-17: Bergmann 2001: desipramine hydrochloride, po, 100 mg, n=6 (AS=2)

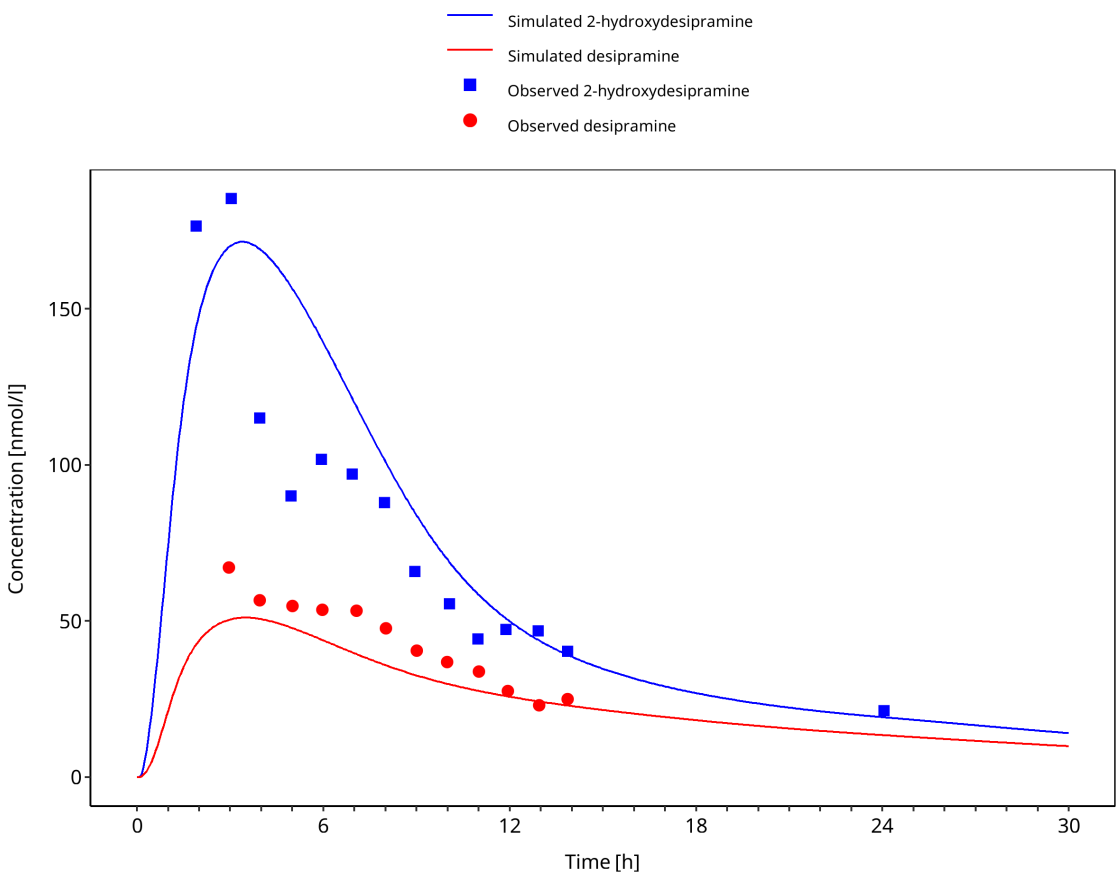


Figure 3-18: Bergmann 2001: desipramine hydrochloride, po, 100 mg, n=6 (AS=2.5)

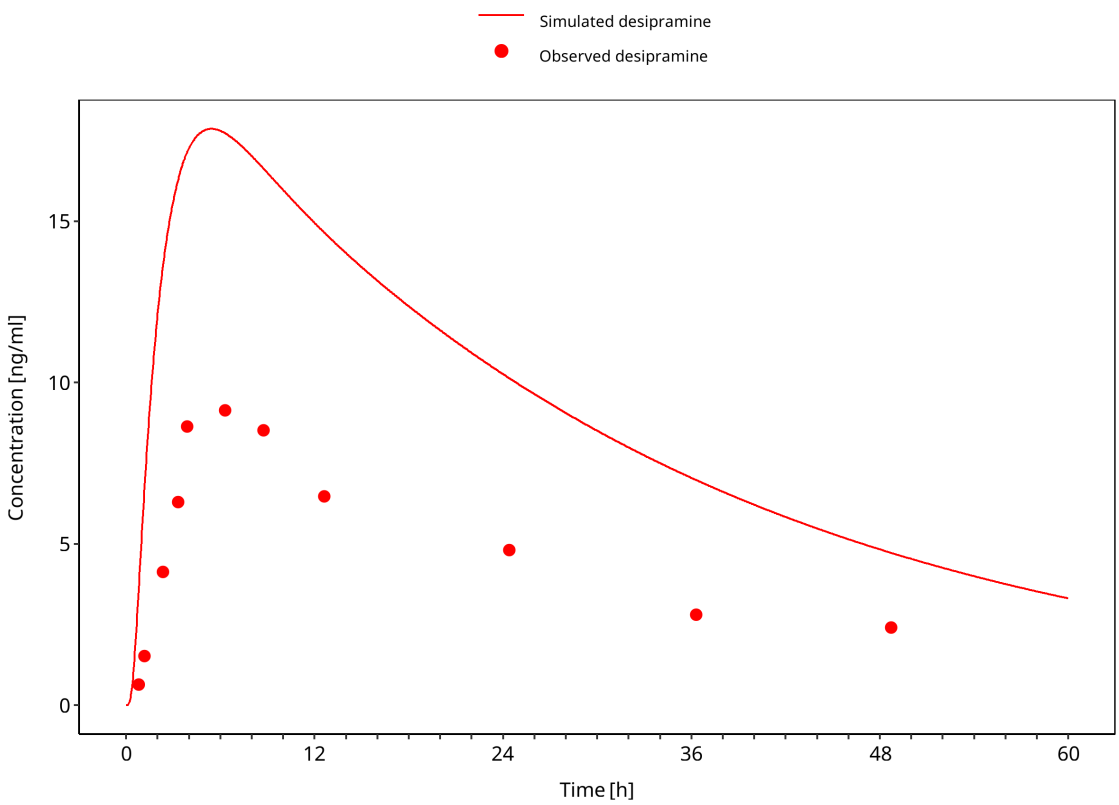


Figure 3-19: Bergstrom 1992: desipramine hydrochloride, po, 50 mg, n=6 (EM)

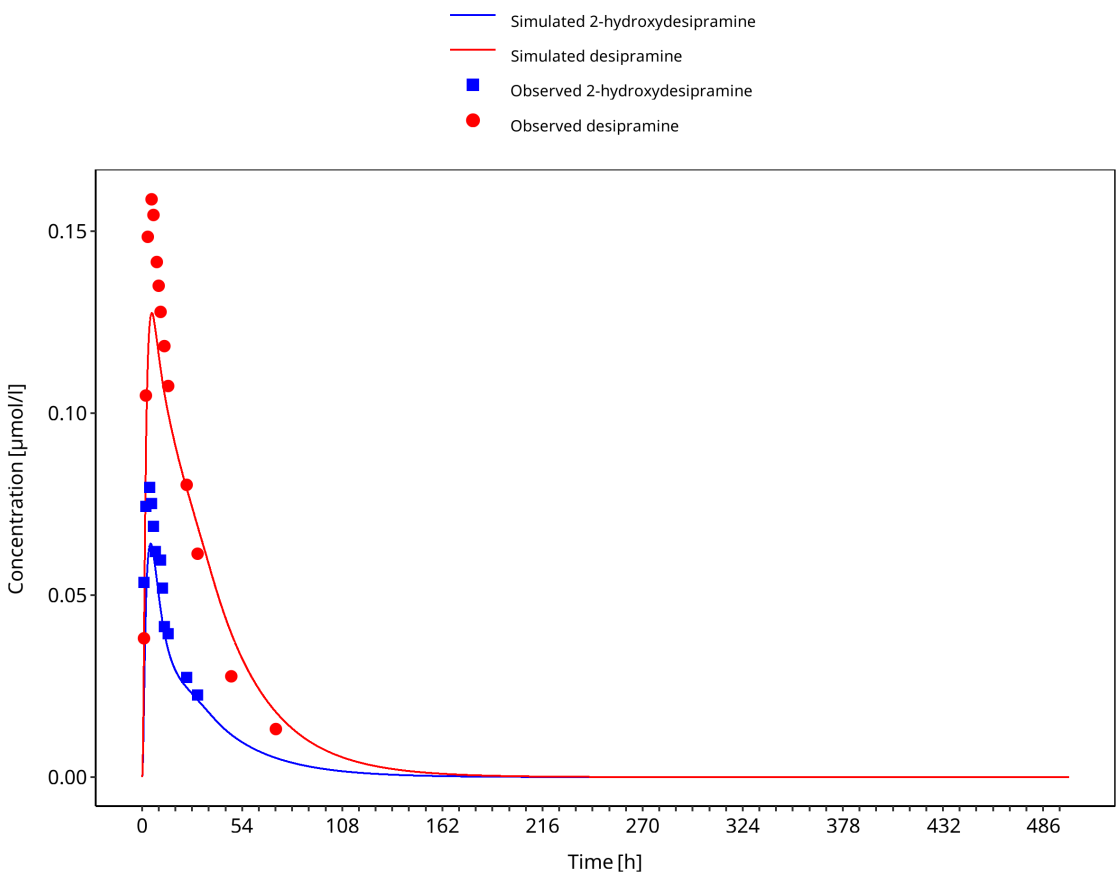


Figure 3-20: Brøsen 1989: desipramine hydrochloride, po, 100 mg, n=6 (EM)

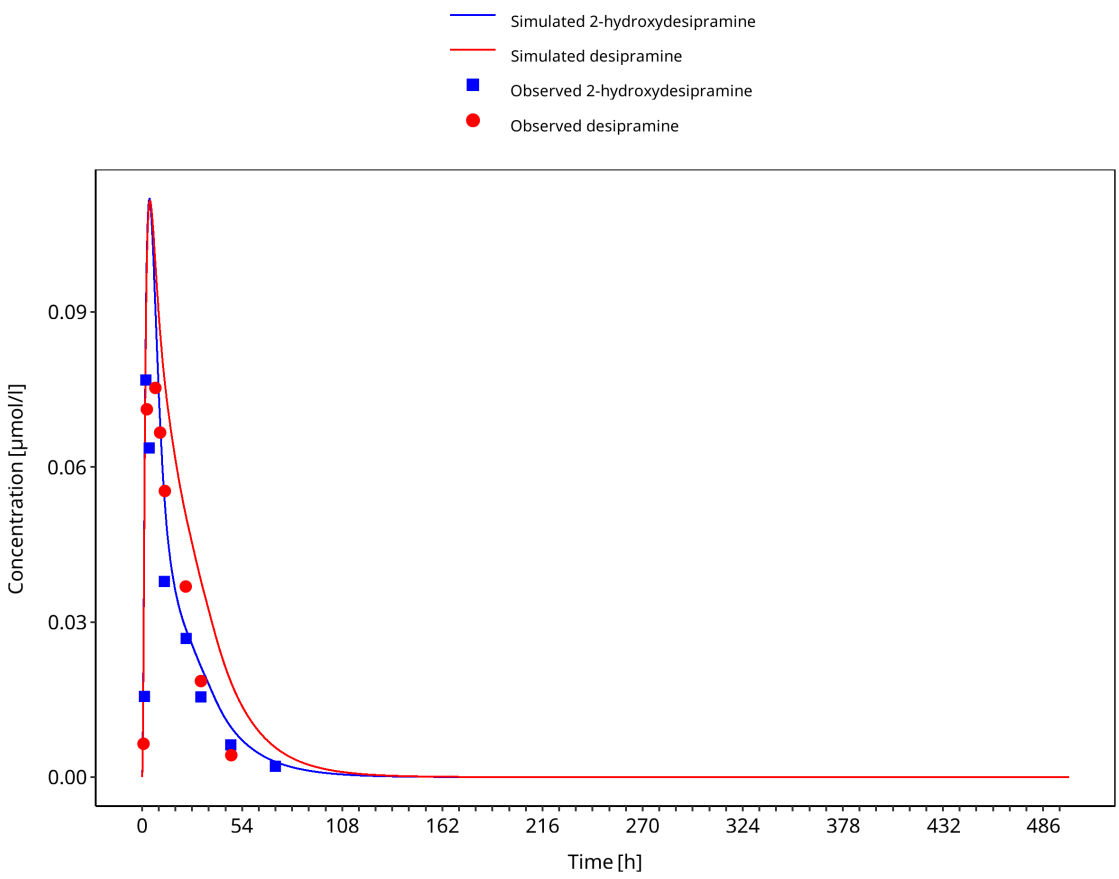


Figure 3-21: Brøsen 1993: desipramine hydrochloride, po, 100 mg, n=4 (fast EM)

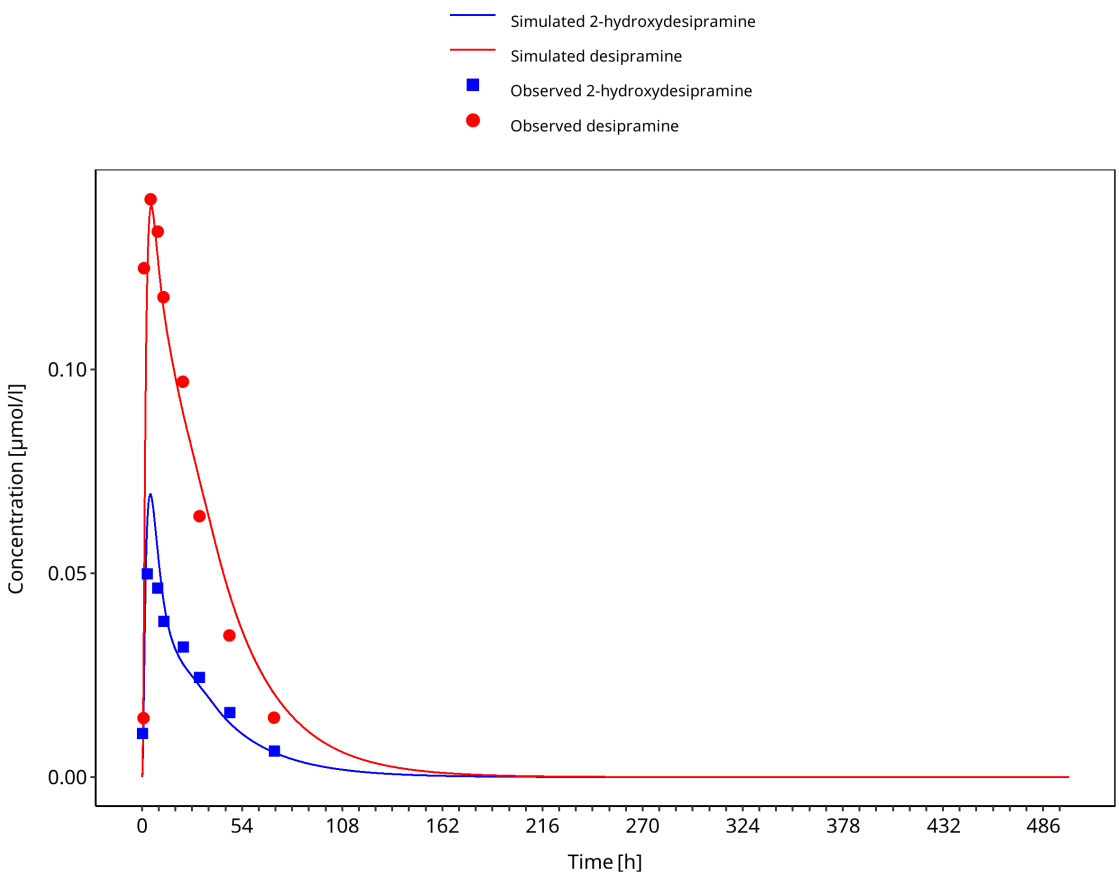


Figure 3-22: Brøsen 1993: desipramine hydrochloride, po, 100 mg, n=5 (EM)

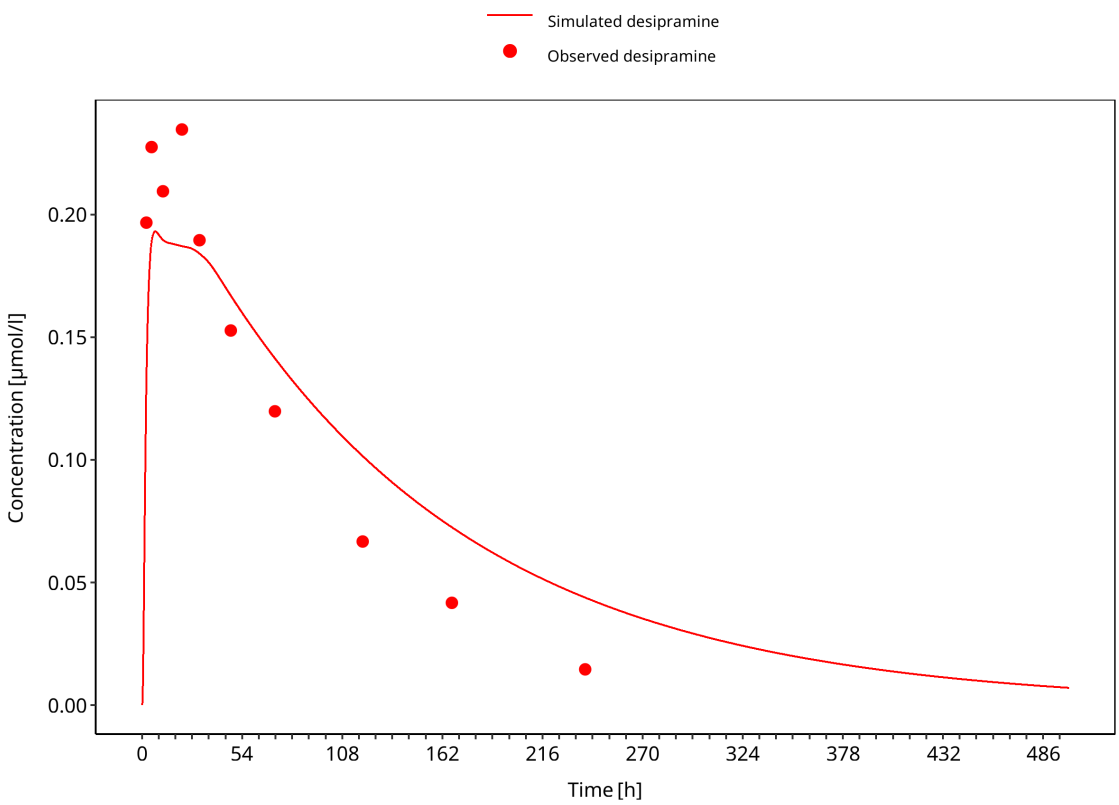


Figure 3-23: Brøsen 1993: desipramine hydrochloride, po, 100 mg, n=8 (PM)

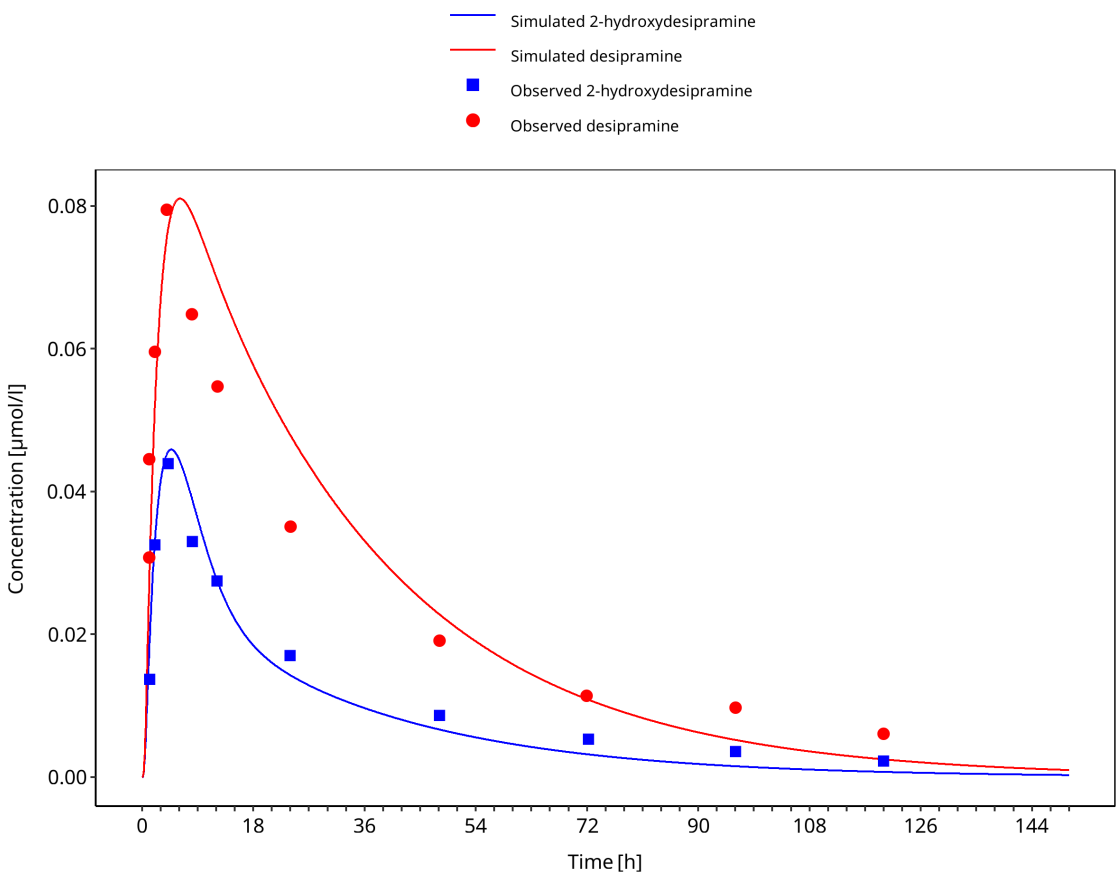


Figure 3-24: Hynes 2015: desipramine hydrochloride, po, 50 mg, n=20 (EM)

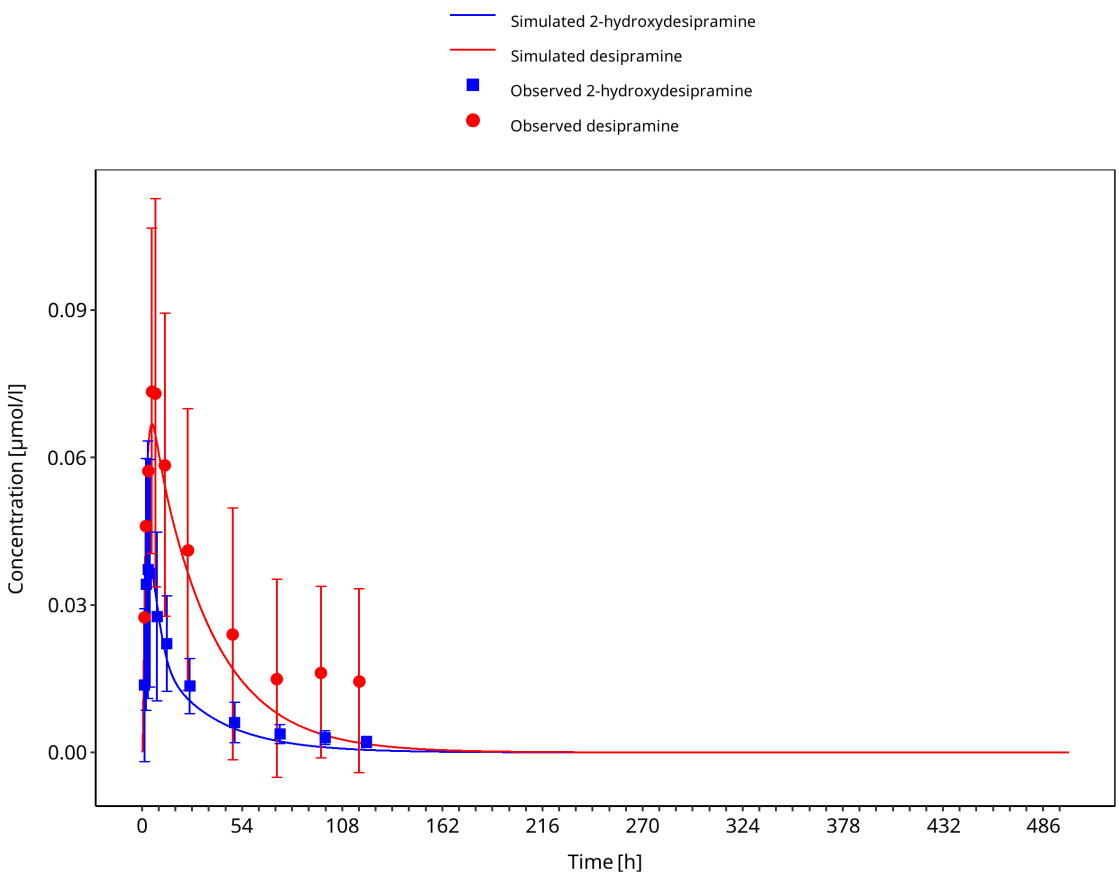


Figure 3-25: Nichols 2009: desipramine hydrochloride, po, 50 mg, n=20 (EM)

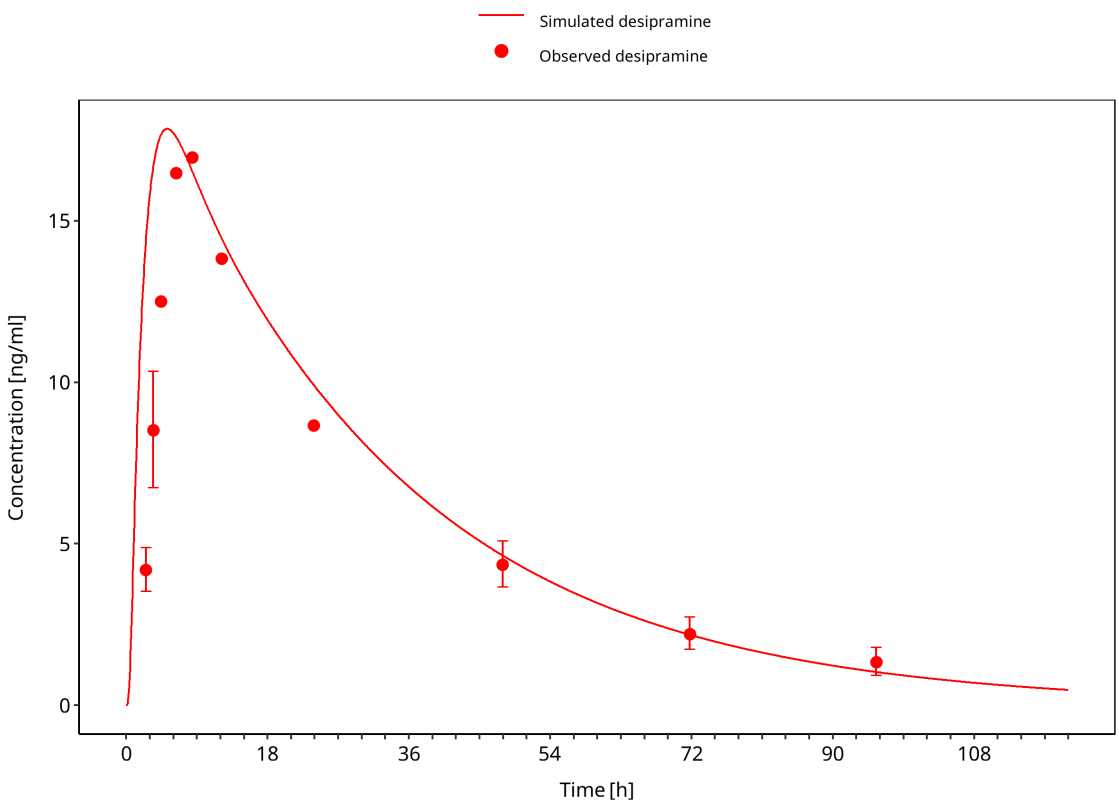


Figure 3-26: Nichols 2013: desipramine hydrochloride, po, 50 mg, n=38 (EM)

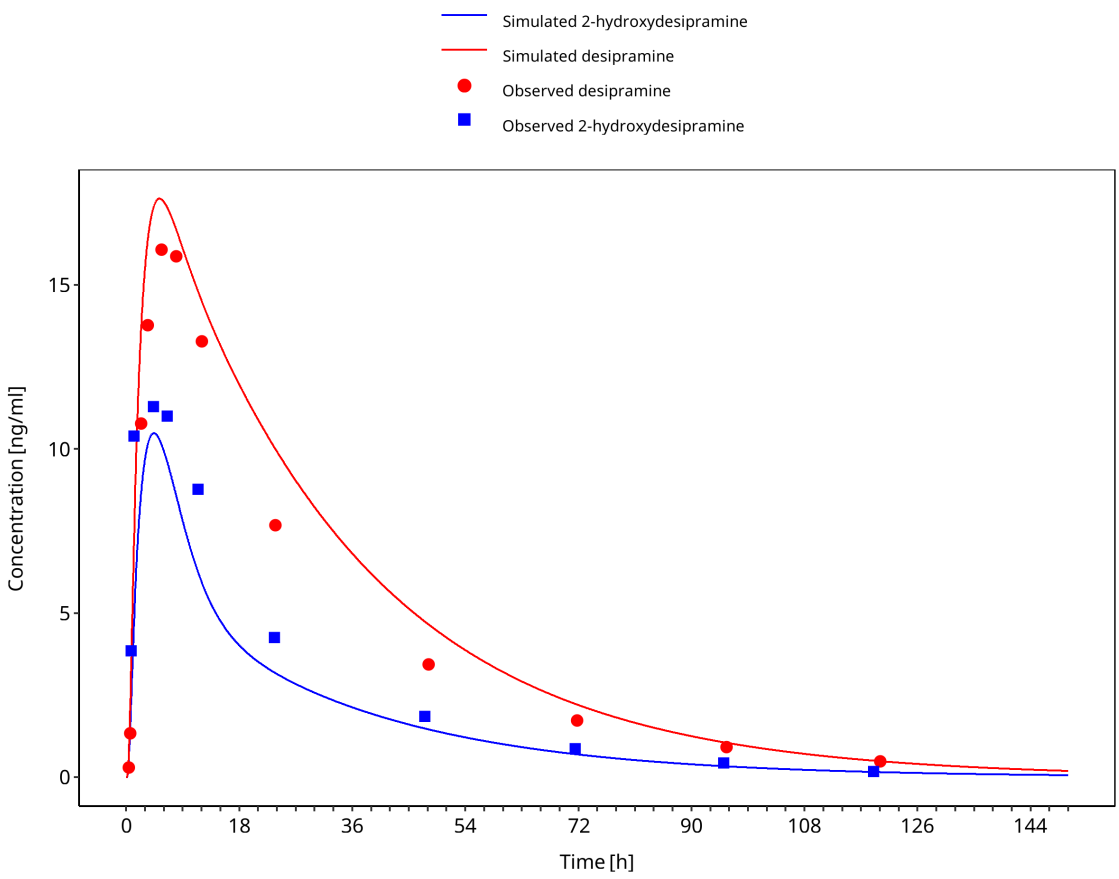


Figure 3-27: Patroneva 2008: desipramine hydrochloride, po, 50 mg, n=20 (EM)

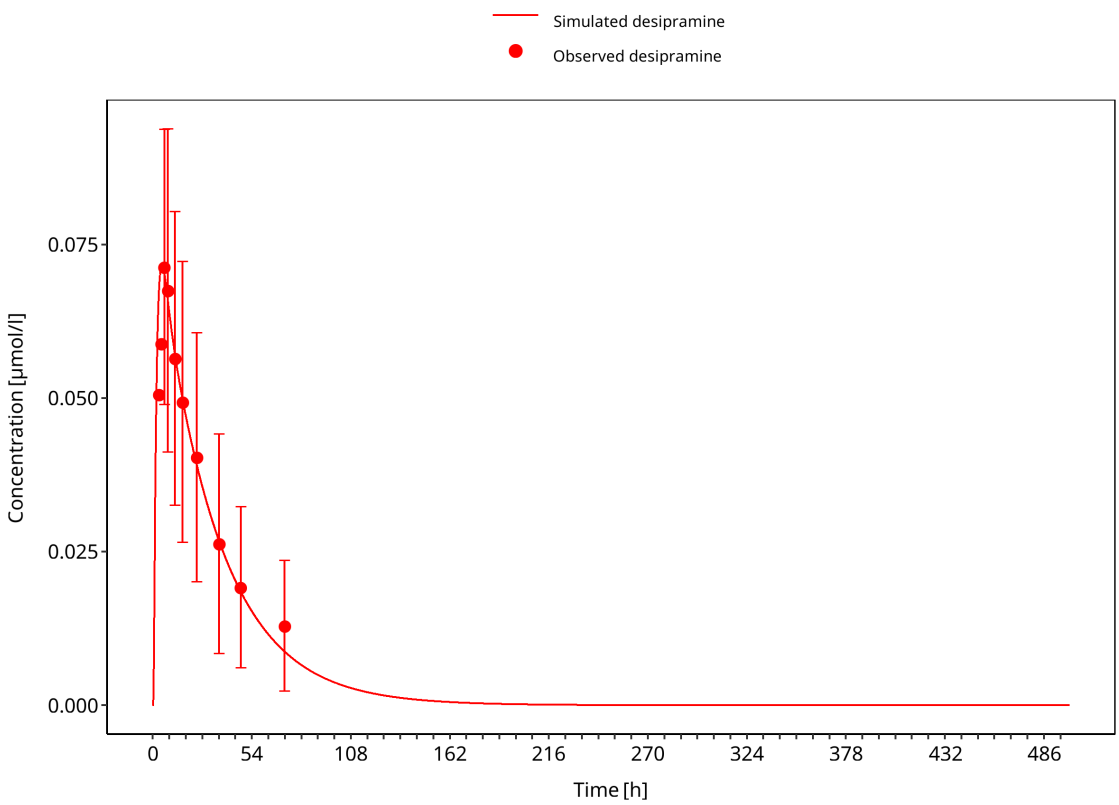


Figure 3-28: Reese 2008: desipramine hydrochloride, po, 50 mg, n=15 (EM)

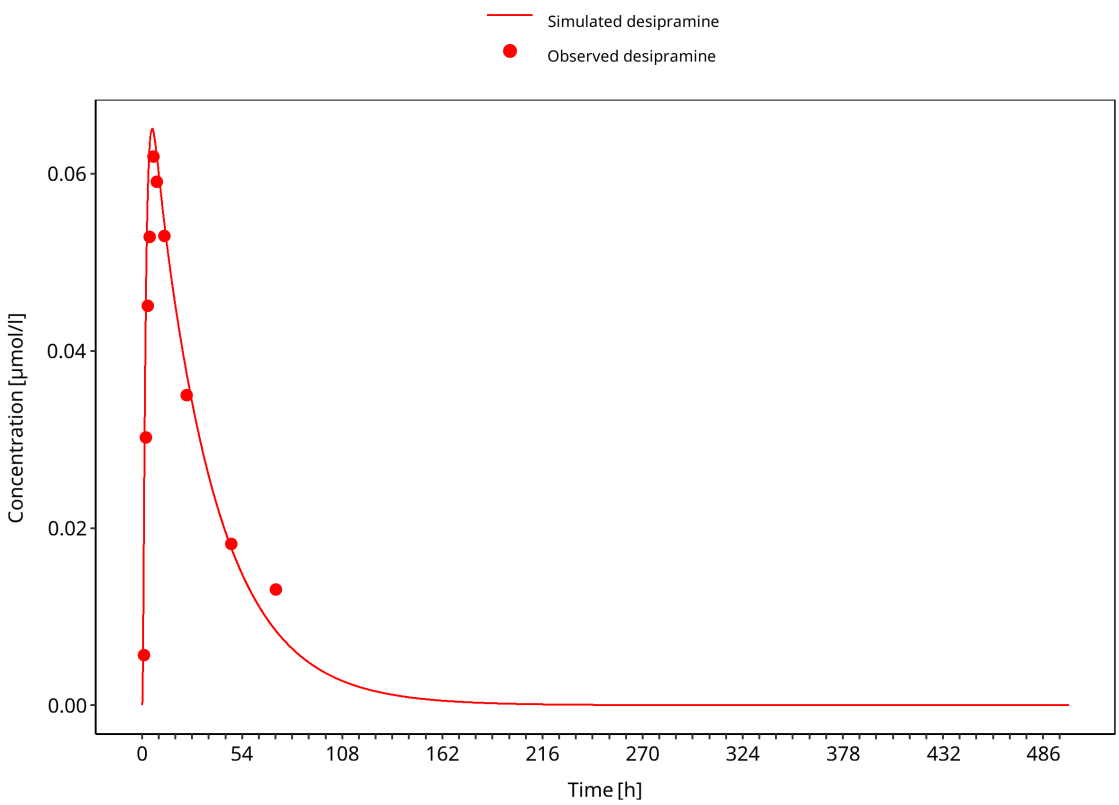


Figure 3-29: Sauer 2004: desipramine hydrochloride, po, 50 mg, n=22 (EM)

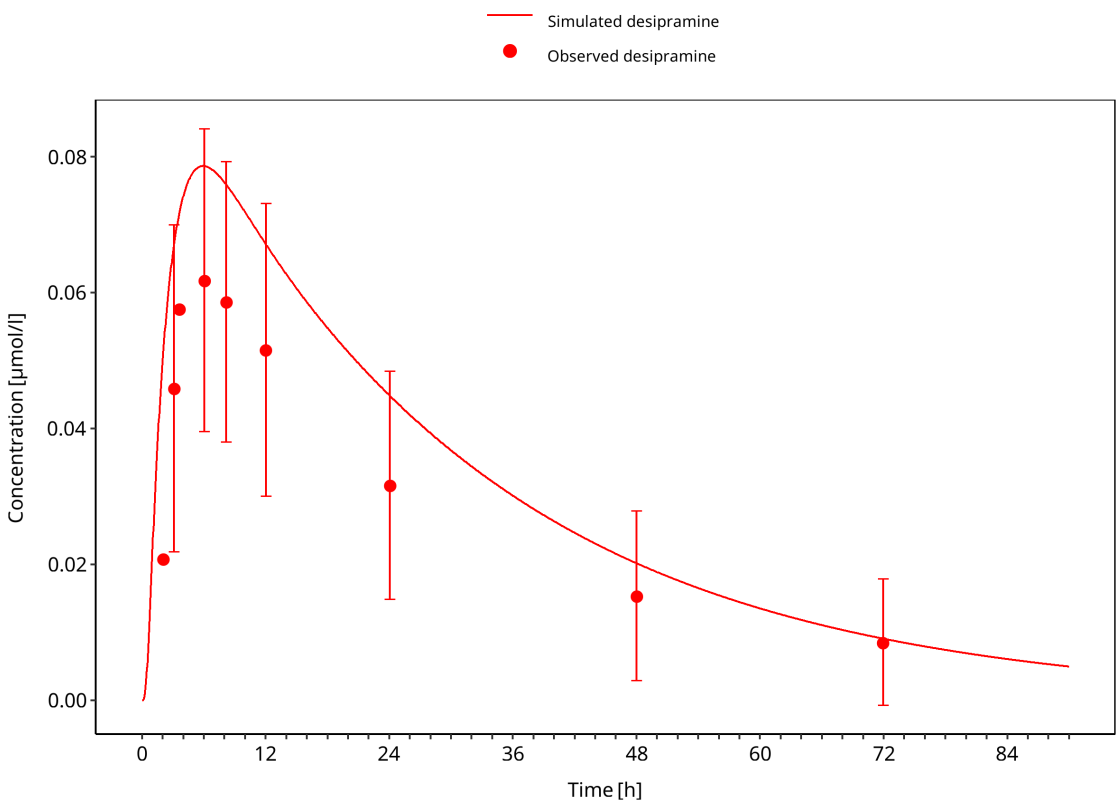


Figure 3-30: Skinner 2003: desipramine hydrochloride, po, 50 mg, n=16 (EM)

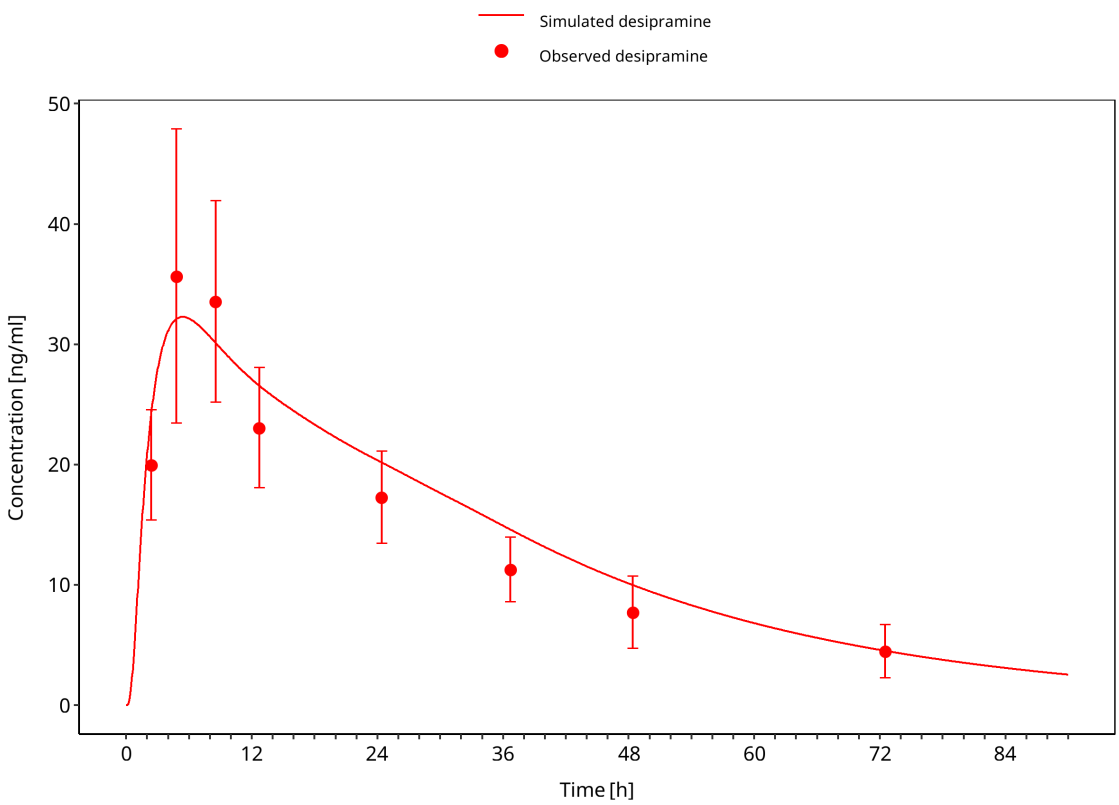


Figure 3-31: Spina 1995: desipramine hydrochloride, po, 100 mg, n=6 (EM)

4 Conclusion

The desipramine PBPK model describes the evaluated plasma concentration-time data for desipramine and 2-hydroxydesipramine after intravenous administration of 50 mg and oral doses of 25-100 mg in adults. The evaluation includes CYP2D6 poor, extensive, normal, and higher-activity groups represented by phenotype or activity score.

The model uses common absorption and distribution parameters across CYP2D6 groups. CYP2D6 forms 2-hydroxydesipramine, while residual hepatic clearance and passive glomerular filtration contribute to parent and metabolite elimination. CYP2D6 characterization-specific catalytic rates describe the observed exposure differences across the evaluated groups.

The concentration-time profiles and goodness-of-fit diagnostics characterize model performance within the evaluated dose, route, analyte, and CYP2D6 ranges. No formal acceptance criterion was applied. This compound report evaluates only desipramine-only or control study arms.

The main limitations are unresolved genotype differences within activity scores, empirical residual clearance, and limited metabolite data. The AS = 2.5 k_{cat} was optimized, and Bergmann 2001 is the only AS = 2.5 data set. The report therefore does not show that this verification profile is independent of the fit.

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